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Cooperative control of a tethered drone and surface vessel for sustained presence at sea

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Thesis to obtain the Master of Science Degree in

Electrical and Computer Engineering

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Declaration

I declare that this document is an original work of my own authorship and that it fulfills all the requirements of the Code of Conduct and Good Practices of the Universidade de Lisboa.

Contents

Acronyms

CG	center of gravity
DDS	Data Distribution Service
DOF	degree of freedom
EMSA	European Maritime Safety Agency
ENU	East-North-Up
LTI	linear time-invariant
MPC	Model Predictive Control
NMPC	Non-linear Model Predictive Control
OCP	optimal control problem
PID	proportional–integral–derivative
ROS	Robot Operating System
SITL	Software-in-the-Loop
UAV	Unmanned Aerial Vehicle
USV	Unmanned Surface Vessel
WAM-V	Wave Adaptive Modular Vessel

Chapter 1

Introduction

This chapter introduces the thesis, presenting the context, motivation, and objectives of the research, followed by an overview of the system architecture, the expected scientific and engineering contributions, and the overall structure of the dissertation.

1.1 Motivation

Saving lives at sea, protecting marine ecosystems, and securing maritime infrastructure all rely on one essential capability: understanding what is happening, where, and when. In vast and dynamic maritime environments, this requires maintaining reliable, high-resolution situational awareness over long periods. The European Maritime Safety Agency (EMSA) recognises this challenge and highlights the need for continuous monitoring and locally persistent operations to enable effective maritime surveillance and response [1].

Different maritime assets meet this challenge in complementary ways. Surface vessels give a physical presence at sea, with long endurance and large payload capacity [2, 3]. However, their range is limited by line of sight, and operations are expensive [4]. Aerial platforms, such as manned aircraft and Unmanned Aerial Vehicles (UAVs), allow fast deployment and elevated sensing [5]. Yet, aircraft are costly, and UAVs suffer from low endurance and unreliable communication [6].

Together, these assets form a connected system rather than separate solutions. Each one compensates for what the other lacks. Unmanned Surface Vessels (USVs) provide long-term presence at sea at much lower cost than crewed vessels, allowing wide and affordable patrols. UAVs extend this reach by sensing beyond the horizon from the air. Working as a team, they maintain continuous and local maritime awareness. This motivates the study of coordinated USV–UAV systems, where cooperation creates mission capabilities that neither platform can achieve alone in complex and changing environments.

A natural way to strengthen this cooperation is to add a physical tether between both platforms. The tether provides continuous power and reliable communication between the USV and UAV [7]. By placing the energy supply and communication systems on the USV, it overcomes two main limits of maritime

UAVs: short endurance and weak long-range communication. This approach has recently transitioned from academic concepts to real-world industrial systems; for instance, the marine robotics company Exail partnered with Elistair to integrate a tethered quadrotor directly onto the DriX USV catamaran for persistent maritime surveillance [8]. Similarly, ACUA Ocean and Teledyne FLIR Defense demonstrated an integrated tethered aerial-surface system for the UK Ministry of Defence to support long-endurance maritime security and target detection [9].

Figure 1.1 illustrates the configuration of the tethered USV–UAV system considered in this work.

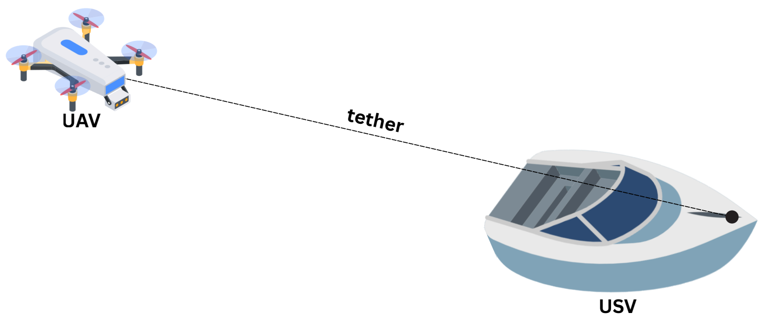


Figure 1.1: Schematic of a tethered USV–UAV system.

The main challenge of this work is to design a control framework that enables two autonomous vehicles, connected by a tether, to operate as a single coordinated system. The controller must manage not only the motion of both vehicles but also account for the tether constraints and dynamics, ensuring safe operation and robustness to wind, waves, and obstacles.

The USV and UAV operate under very different dynamics. The USV moves slowly, dominated by hydrodynamic drag and inertia, while the UAV reacts quickly, with motion evolving on much shorter time scales. Coordinated control is challenging because the controller must handle these different response rates and actuation dynamics consistently.

The maritime environment adds further complexity. Wind, waves, and water motion constantly disturb both platforms, making robustness to disturbances essential. Nearby structures, vessels, or obstacles can restrict their movement and risk tether entanglement. Effective cooperation relies on continuous information exchange and synchronised decision-making between both agents.

Most existing tethered UAV systems focus on power delivery and communication, with operation often remaining manual or semi-autonomous. In research prototypes, the aerial and surface vehicles are usually decoupled, the tether is simplified or linearised, or the surface platform is treated as fixed. These assumptions make control design easier but reduce realism. Few cooperative control approaches explicitly model the tether while coordinating both agents under realistic maritime conditions.

1.2 Problem Statement

This work addresses the problem of controlling a cooperative system composed of an autonomous surface vessel (the WAM-V USV) and a physically connected quadrotor (the x500 UAV) operating in a coordinated manner. The objective of such a system is to autonomously execute maritime missions by following predefined waypoints or continuous reference trajectories, while respecting the physical and operational constraints of both platforms and of the tether that connects them.

To achieve this objective, the control framework must be capable of handling different phases of operation and mission requirements. This includes the definition of multiple control modes that allow the system to perform manoeuvres such as takeoff and landing, as well as different forms of coordinated motion during navigation. The UAV NMPC must actively prevent the tether from exceeding its maximum physical length, utilizing soft constraints and slacks, while the USV NMPC coordinates the planar movement of the dynamic anchor deck.

1.3 System Overview

Interaction with the system is defined at the mission level. The operator selects a mission, such as patrolling a given area, following a moving target, or visiting a set of waypoints that represent regions of interest. This mission information is received by a behaviour management layer, which selects the active control mode and handles mission logic, including switching between different relative configurations of the UAV with respect to the USV, as well as executing specific manoeuvres such as takeoff and landing.

Based on the selected mission and active control mode, the system generates coordinated motion for both the USV and the UAV. This process combines planning and control to compute trajectories and actuation commands that satisfy vehicle and tether constraints while pursuing the mission objectives. The WAM-V USV acts as the leader planar platform carrying an active winch system on its deck. The winch dynamically spools the tether in or out to manage slack and cable tension. The x500 UAV follows a relative orbit or observation trajectory above the USV, using its elevated perspective to record visual data.

Figure 1.2 presents a high-level block diagram of the proposed system, showing the interaction between the coordinator, the decentralized NMPC solvers, the vehicles, and the co-simulation interfaces.

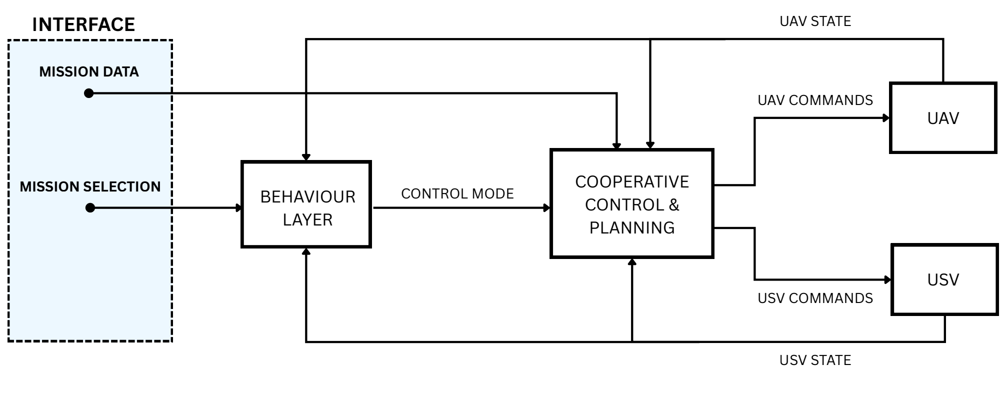


Figure 1.2: High-level block diagram of the proposed cooperative control system.

1.4 Expected Contributions

This thesis contributes to the design, implementation, and evaluation of cooperative control solutions for tethered UAV–USV systems, to enable autonomous and persistent maritime missions. The main contributions of this work are summarised as follows:

- A unified, C++ ROS 2 control pipeline utilizing Acados and HPIPM to generate and execute real-time Nonlinear Model Predictive Control solvers for both surface tracking and aerial guidance.
- The modeling and experimental identification of the UAV inner-loop attitude response lag ($\tau = 0.15\,\mathrm{s}$) using PX4 doublets flight data.
- The design and implementation of a virtual winch controller operating in both geometric and tension-feedback PD modes to prevent cable slackness and tension spikes.
- A realistic co-simulation setup in Gazebo Harmonic, integrated with the PX4 Software-in-the-Loop (SITL) firmware and the MoorDyn physics engine for multi-body lumped-mass line representation.
- The verification of NMPC constraints enforcement (states, inputs, and tether length limits) and evaluation of solver latency.
- The validation of the entire cooperative control stack under an integrated search and pursuit mission targeting a vessel teleoperated in real-time.

1.5 Thesis Structure

The remainder of this dissertation is organized as follows:

- **Chapter 2 (Literature Review and State of the Art)** reviews existing research in cooperative marine-aerial robotics, tethered drones, and predictive control algorithms.
- **Chapter 3 (System Modeling)** derives the kinematic and dynamic models of both vehicles, as well as the coupling tether and winch formulations.
- **Chapter 4 (Cooperative Control Design)** details the decentralized NMPC mathematical formulation, cost functions, physical constraints, and winch spooling control laws.
- **Chapter 5 (Simulation Environment and Software Implementation)** introduces the physical characteristics of the virtual WAM-V and x500 platforms, the simulation setup, and the software modularity.
- **Chapter 6 (Results and Discussion)** evaluates the model validation, parameter identification, tracking precision, constraint enforcement, solver latency, and the target pursuit mission.
- **Chapter 7 (Conclusions and Future Work)** summarizes the findings of the thesis and outlines directions for future research.

Chapter 2

State of the Art and Related Work

This chapter is structured into four main sections, each representing a key element of cooperative system control. Each section presents the main techniques commonly used in control theory, together with subsections reviewing related work on topics closely related to this research.

The first section addresses *modelling and identification*, which provides the mathematical and experimental tools required to describe system dynamics and estimate relevant parameters, including tether and cable physics. The second focuses on *control strategies*, ensuring stability, accurate tracking, and robustness to disturbances at the individual agent level, including predictive control and active winch architectures. The third section concerns *cooperative coordination*, which enables communication and shared decision-making so that multiple agents can operate as a unified system. Finally, *path and trajectory planning* deals with the generation of feasible and safe motion plans that connect mission objectives with low-level control actions.

2.1 Modeling and System Identification

Understanding how a system evolves is a prerequisite for any control design. Modelling provides this understanding by offering a mathematical description of the system dynamics, capturing the dominant behaviours while deliberately neglecting secondary effects that are not essential for analysis or control purposes [10]. The level of detail of such a model is inherently linked to the available physical insight, the amount and quality of experimental data, and the specific objectives for which the model is intended.

2.1.1 White-Box, Grey-Box, and Black-Box Modeling

Depending on the available physical knowledge and data, modeling methodologies are classified into three main groups:

- **White-Box Modelling:** Based on first principles and physical laws that govern system behaviour. By expressing the dynamics through analytical formulations, often derived from Newtonian mechanics, hydrodynamics, or aerodynamics, this approach provides clear insight into stability, controllability, and energy transfer. To preserve analytical tractability, simplifying assumptions such as linearisation or the neglect of higher-order effects are commonly introduced.

- **Grey-Box Modelling:** Lies between purely analytical and purely data-driven approaches. The structure of the model is defined using physical insight, while parameters that are difficult to determine analytically, such as hydrodynamic or aerodynamic coefficients, are estimated from experimental data. This combination preserves a clear physical interpretation while improving accuracy through calibration.

- **Black-Box Modelling:** Relying entirely on measured input–output data to describe system behaviour. Rather than using explicit physical laws, this approach identifies mathematical relationships that best fit the observed data. Typical examples include linear parametric models, state-space representations obtained from data, and learning-based methods.

2.1.2 Tether and Cable Dynamics Modeling

Modeling flexible tethers is critical since they introduce highly nonlinear forces and kinematic constraints on the vehicles:

- **Quasistatic Catenary Models:** Describe the steady-state profile of a heavy cable suspended between two points. These models use catenary equations to resolve cable sag and tension analytically, assuming the line is in static equilibrium at each step. While computationally efficient, they neglect dynamic effects such as transverse wave propagation and drag forces, making them unsuitable for rapid aerial maneuvers.

- **Lumped-Mass Models:** Discretize the tether into a finite chain of nodes, or "lumped masses", connected by spring-damper segments. Each segment models axial elasticity, damping, and hydrodynamic drag, while the nodes resolve gravitational and buoyancy forces. This formulation, implemented in solvers such as MoorDyn, offers a high-fidelity representation of transient tether whip, slackness, and hydrodynamic drag under dynamic boundary conditions.

- **Finite-Segment and Multi-Body Models:** Formulate the line as a chain of rigid cylinders connected by spherical joints. This approach computes bending and torsional moments accurately, but requires solving high-index differential-algebraic equations, which increases the computational burden.

2.1.3 Optimization-Based Estimation

System identification offers an alternative path by constructing models directly from measured input–output data when physical knowledge is incomplete. Optimisation-based estimation focuses on identifying model parameters offline by minimising the mismatch between measured and predicted system outputs. This is typically achieved through cost functions based on least-squares formulations, including weighted least-squares and prediction-error methods (PEM) [11]. For nonlinear or constrained problems, advanced optimisation schemes such as trust-region methods and constrained least-squares are often employed, allowing grey-box models to be calibrated while enforcing physical bounds and regularisation.

2.1.4 Related Work on System Identification

In systems connected by a tether, the model of the overall system must account at least for the tether geometry to constrain the relative motion of the agents and avoid physically infeasible configurations that could lead to tether failure.

In [12], a buoy–UAV system is presented in which the tether is modelled explicitly as a geometric and dynamic constraint between the buoy and the aerial vehicle. The tether tension is computed from the coupled dynamics and incorporated as an external force acting along the tether direction in the UAV model. This formulation relies on simplifying assumptions, namely that the tether is inextensible, massless, free of hydrodynamic effects, and remains taut during coupled operation.

An alternative modelling perspective is proposed in [13], where tether-induced forces are not included explicitly in the system dynamics. Instead, their effect is lumped into an external disturbance acting on the UAV. This disturbance is then handled at the control level through active disturbance rejection, avoiding the need for explicit tether modelling while still compensating for its influence in real time.

In [14], a cooperative UAV–USV system is modelled for the purpose of manipulating a floating object. Both vehicles are connected to the object through individual tethers, and the full system is described using simplified white-box models. The formulation assumes inextensible and massless tethers that remain permanently taut.

In [15], a tether is not included, and both the UAV and the USV are described using simplified white-box models. The USV is represented by a planar marine vehicle model commonly used for path-following applications, while the UAV is modelled independently using a reduced-order dynamic model.

2.2 Control Strategies for Autonomous Vehicles

Control theory provides the mathematical framework for designing systems that regulate their behaviour through feedback, ensuring stability, accuracy, and robustness in the presence of uncertainty and disturbances.

2.2.1 Classical and Modern Control

Classical control is based on the representation of systems through transfer functions in the Laplace or frequency domain, under the assumption of linear time-invariant (linear time-invariant (LTI)) dynamics. Within this framework, feedback laws are typically implemented through proportional–integral–derivative (proportional–integral–derivative (PID)) controllers and frequency-domain compensators designed to shape the closed-loop response.

Modern control theory generalises these principles to systems that are multivariable, time-varying, or subject to strong coupling between states and inputs. It is formulated in the *state-space* domain, where the internal dynamics of a system are described by sets of first-order differential equations. This representation enables systematic analysis of controllability, observability, and stability. Within this framework, optimal control theory defines feedback laws that minimise a performance index, such as the linear quadratic regulator (LQR) or robust control methods such as H_∞ [16, 17].

2.2.2 Predictive and Optimization-Based Control

Predictive and optimisation-based control methods determine control actions by solving an optimisation problem that predicts the system behaviour over a finite horizon. At each sampling instant, the controller minimises a cost function subject to system dynamics and operational constraints, typically penalising deviations from reference trajectories and excessive control effort. Only the first control action is applied, and the process repeats at the next time step, forming the receding-horizon principle that defines Model Predictive Control (MPC) [18].

Extensions to nonlinear systems led to the development of Non-linear Model Predictive Control (NMPC), which replaces linear models with nonlinear formulations and uses numerical optimisation techniques to compute control inputs in real time [19, 20]. To enable real-time embedded execution on high-rate systems (e.g. UAV flight control), modern applications rely on symbolic framework code generators (such as CasADi) that export native C solvers. These solvers utilize highly efficient quadratic programming kernels such as HPIPM and execution frameworks like the Sequential Quadratic Programming Real-Time Iteration (SQP-RTI) scheme, which performs a single optimization iteration per time step to minimize solver latency.

2.2.3 Active Winch and Tension Control

In tethered UAV operations, winch design is a core element:

- **Passive Tension Tethers:** Rely on spring-loaded spools or constant-torque slip rings. While simple and requiring no electrical control, they cannot adapt to fast vessel maneuvers, leading to large tension spikes or excessive slack.
- **Active Spooling Winches:** Utilize motorized actuators regulated by microcontrollers. The winch dynamically adjusts the tether length based on geometric tracking (distance between platforms) or force-feedback algorithms.
- **Tension-Feedback and Active Heave Compensation (AHC):** Read force telemetry from a load cell. A closed-loop proportional-derivative (PD) controller regulates the winch spooling speed to maintain a target tension. In maritime environments, this functions as AHC, isolating the flying quadrotor from wave-induced vessel heave.

2.2.4 Related Work on Tethered Vehicle Control

In tethered UAV systems, the tether introduces additional forces and coupling effects that influence control design. In [12], this coupling is addressed through a nonlinear backstepping controller derived directly from the white-box dynamics of the UAV–buoy system. The work in [13] adopts an alternative approach based on active disturbance rejection, where the tether influence is not explicitly modelled but treated as an external disturbance estimated in real time by a reduced-order observer.

In [14], cooperative manipulation with two tethered agents is controlled using a MPC formulation based on simplified white-box models of the UAV, the USV, and the floating object. A similar predictive strategy is presented in [21], where both the UAV and the USV are coordinated using NMPC, relying on known dynamic models.

In [15], the USV employs a line-of-sight guidance law combined with PID control for heading and speed regulation. The UAV motion is controlled using an NMPC formulation, which generates reference accelerations to track the desired relative trajectory. No joint or centralised control law is employed, and each vehicle is controlled using its own dedicated controller.

2.3 Cooperative Coordination and Systems

Cooperative coordination refers to the control and organisation of multiple agents working towards achieving goals through communication and joint decision-making.

2.3.1 Centralized vs. Distributed Coordination

Depending on how information is exchanged and the control authority is structured, cooperative architectures are broadly categorised as centralized or distributed:

- **Centralized Coordination:** A single decision node aggregates global information and computes control commands for all agents. This architecture facilitates global optimisation and ensures consistency of mission-level objectives, since the planner has access to full system states and constraints. However, centralized control scales poorly with team size and is vulnerable to communication delays or single-point failures.

- **Distributed Coordination:** Eliminates the need for a central planner by allowing each agent to make decisions based on locally available information and limited communication. Agents exchange state or intent data to achieve group-level objectives. Distributed coordination provides scalability and robustness to communication losses and agent failures, though it faces challenges related to synchronization and network stability.

2.3.2 UAV-USV Cooperative Systems

Cooperative deployments of UAVs and USVs combine the speed and altitude of aerial sensing with the weight-bearing capability and range of marine hulls:

- **Autonomous Landing on Moving Surfaces:** Focuses on landing a quadrotor on a moving deck (helideck) in waves. This requires precise relative state estimation (visual markers, RTK-GPS) and predictive control to synchronize the UAV's landing velocity with the sea-state pitch/roll profile of the USV deck.

- **Mobile Recharging Stations:** The USV acts as a mother ship. Drones fly reconnaissance sweeps, return to land on the USV deck to swap or recharge batteries, and take off again, enabling long-term persistence at sea.
- **Cooperative Target Tracking:** Cooperative path-following algorithms steer the USV planar trajectory to intercept a target, while the UAV orbits above to provide continuous visual tracking and image feedback.

2.3.3 Related Work on Maritime Cooperative Robotics

In [21], cooperation is based on a tracking strategy in which the UAV is controlled to follow and land on a moving USV. Only the UAV is actively controlled, while the USV moves independently and provides its state as a reference. The interaction is therefore unidirectional, with the UAV adapting its motion to the USV.

In contrast, [14] proposes a centralised model predictive control approach in which the UAV and the USV are controlled jointly. A single optimisation problem computes the control inputs for both vehicles, explicitly accounting for the coupling introduced by the tether. This approach focuses on cooperative object manipulation rather than on mission-level coordination between the vehicles.

In [15], coordination is achieved through a leader–follower structure. The USV acts as the leader and follows a predefined path, while the UAV adjusts its motion to maintain a desired relative configuration with respect to the USV. Coordination is therefore based on reference sharing and relative motion constraints, without joint optimisation or centralised decision-making.

2.4 Path and Trajectory Planning

Path and trajectory planning address the problem of generating feasible, collision-free, and dynamically consistent motions for autonomous systems operating in structured or unstructured environments, translating abstract goals into trackable trajectories.

2.4.1 Offline and Online Planning

Methods are broadly divided into offline and online approaches, depending on when the plan is computed and how environmental information is handled:

- **Offline Planning:** Performed before execution, assuming full or near-complete knowledge of the operating environment. Classical approaches include graph-search methods (Dijkstra's, A*), sampling-based methods (RRT, PRM), and trajectory optimization. While generating globally optimal paths, they are computationally demanding and cannot adapt to dynamic changes.

- **Online Planning:** Addresses the generation and adaptation of trajectories in real time under dynamic or uncertain conditions. Reactive techniques such as potential field methods and dynamic window approaches provide fast obstacle avoidance but can suffer from local minima. Optimization-based MPC frameworks integrate vehicle dynamics into the planning loop, generating dynamically consistent collision-free paths online.

2.4.2 Related Work on Trajectory Planning

A centralised predictive planning approach is presented in [14], where a single MPC formulation jointly optimises the motion of both the UAV and the USV. In this case, planning is achieved online through joint trajectory optimisation that explicitly accounts for the coupling introduced by tethers.

Planning is addressed from a different perspective in [15], where cooperative behaviour is achieved through coordinated path following. A predefined path is assigned to the USV, while the UAV generates its motion online as part of the control process to maintain coordination. In this framework, planning is limited to the online generation of the UAV motion, and no cooperative path planning between the two vehicles is performed.

Overall, existing works typically adopt one of two approaches to planning in UAV–USV systems, either a predefined path is assigned to one vehicle while the other generates its motion online to follow or support it, or both vehicles generate their motion online as part of a predictive control framework.

Chapter 3

System modeling

Real-time control requires simplified yet representative models that describe the motion and physical interaction of each vehicle. To this end, this chapter describes the kinematics, dynamics, and physical coupling of the cooperative system. These formulations allow the controller to predict future states and compute feasible, optimal trajectories for the system.

First, the chapter defines the reference coordinate frames and spatial transformations. It then develops the kinematic and dynamic models for the USV and the UAV. Finally, it establishes the geometric and force models for the flexible tether and active winch.

3.1 Coordinate frames and transformations

Formulating the equations of motion for the cooperative system requires a consistent set of coordinate frames. To this end, this work defines a common global inertial reference frame alongside individual body-fixed frames for both the USV and the UAV. This formulation simplifies the representation of hydrodynamic and aerodynamic forces relative to each vehicle. Figure 3.1 illustrates the spatial relationships between these coordinate systems.

3.1.1 Reference coordinate frames

Inertial Reference Frame $\{\mathcal{I}\}$ This frame establishes the global coordinate system, following the East-North-Up (ENU) convention. Its origin $\mathcal{O}_I$ lies on the water surface. In this frame, the axis $\mathbf{x}_I$ points East, $\mathbf{y}_I$ points North, and $\mathbf{z}_I$ points upward, opposite to gravity. This flat-Earth assumption is justified because the local tracking controller operates over a short spatial horizon.

USV Body-Fixed Reference Frame $\{\mathcal{B}_v\}$ This frame attaches directly to the USV. Its origin $\mathcal{O}_v$ lies at the USV's center of gravity, a placement that simplifies the equations of motion by decoupling linear and rotational movements. Regarding the axes, the longitudinal axis $\mathbf{x}_v$ points forward (surge), the lateral axis $\mathbf{y}_v$ points to port (sway, left), and the vertical axis $\mathbf{z}_v$ points upward (heave).

UAV Body-Fixed Reference Frame $\{\mathcal{B}_a\}$ Similarly, this frame attaches to the UAV. Its origin $\mathcal{O}_a$ resides at the UAV's center of gravity. The longitudinal axis $\mathbf{x}_a$ points forward, the lateral axis $\mathbf{y}_a$ points to the left, and the vertical axis $\mathbf{z}_a$ points upward, aligning with the collective thrust.

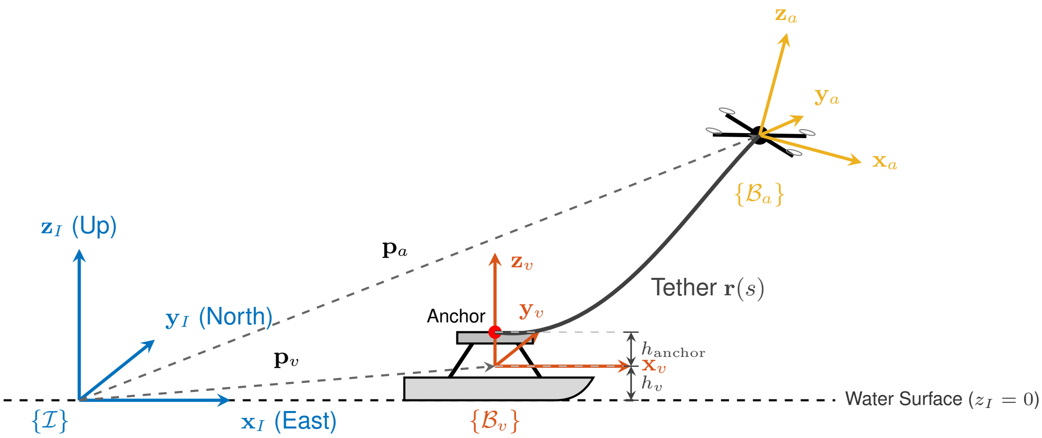


Figure 3.1: Reference coordinate frames $\{\mathcal{I}\}$, $\{\mathcal{B}_v\}$, $\{\mathcal{B}_a\}$, and anchor connection.

3.1.2 Coordinate transformations

To represent the physical locations of both vehicles relative to the global coordinate system, the model defines their position vectors in the inertial frame $\{\mathcal{I}\}$. The vector $\mathbf{p}_v = [x_v, y_v, h_v]^T$ defines the position of the USV's center of gravity, where the model assumes a constant altitude h_v above the water surface to simplify the motion equations by neglecting heave dynamics. Similarly, the vector $\mathbf{p}_a = [x_a, y_a, z_a]^T$ represents the position of the UAV's center of gravity.

Projecting coordinates, velocities, and forces from the body-fixed frame $\{\mathcal{B}_v\}$ to the inertial frame $\{\mathcal{I}\}$ requires a rotation matrix. Given that the USV operates on the water surface with horizontal propulsion, and its inherent stability minimizes roll and pitch, the model assumes planar motion. Under this assumption, the USV rotation matrix $\mathbf{R}_v(\psi)$ performs this projection depending only on the yaw angle ψ:

$$\mathbf{R}_v(\psi) = \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{3.1}$$

Similarly, projecting the UAV's thrust force from its body-fixed frame $\{\mathcal{B}_a\}$ to the global inertial frame $\{\mathcal{I}\}$ requires a rotation matrix $\mathbf{R}_a(\phi, \theta, \psi_a)$. Parameterized by the roll (ϕ), pitch (θ), and yaw (ψ_a) angles, this matrix follows the standard ZYX rotation sequence:

$$\mathbf{R}_a(\phi, \theta, \psi_a) = \begin{bmatrix} \cos\psi_a\cos\theta & \cos\psi_a\sin\theta\sin\phi - \sin\psi_a\cos\phi & \cos\psi_a\sin\theta\cos\phi + \sin\psi_a\sin\phi \\ \sin\psi_a\cos\theta & \sin\psi_a\sin\theta\sin\phi + \cos\psi_a\cos\phi & \sin\psi_a\sin\theta\cos\phi - \cos\psi_a\sin\phi \\ -\sin\theta & \cos\theta\sin\phi & \cos\theta\cos\phi \end{bmatrix} \tag{3.2}$$

✓

In the USV, the tether is attached to a winch located at a vertical offset $\mathbf{r}_{\text{anchor}} = [0, 0, h_{\text{anchor}}]^T$ relative to the USV's CG in $\{\mathcal{B}_v\}$. The vertical component h_{anchor} represents the constant height of the anchor point above the CG. Projecting this offset into the global frame $\{\mathcal{I}\}$ yields the global anchor position:

$$\mathbf{p}_{\text{anchor}} = \begin{bmatrix} x_v \\ y_v \\ h_v \end{bmatrix} + \mathbf{r}_{\text{anchor}} = \begin{bmatrix} x_v \\ y_v \\ h_v + h_{\text{anchor}} \end{bmatrix} \tag{3.3}$$

The tether attachment point on the UAV is assumed to coincide exactly with its center of gravity. This simplification eliminates any offset in the UAV's body frame $\{\mathcal{B}_a\}$, meaning the tether tension force acts directly on the CG and does not introduce attitude coupling moments. Consequently, the global position of the UAV's tether connection is equivalent to the UAV's position. ✓

3.2 WAM-V Surface Vessel Model

- **Objective:** Formulate the kinematic and dynamic equations of motion for the WAM-V (Wave Adaptive Modular Vessel) catamaran platform for real-time optimal control.
- **Modeling Hierarchy:**
  - Planar 3-degree of freedom (DOF) kinematics on the horizontal sea surface.
  - Control-oriented non-linear dynamic model based on Fossen's marine craft formulation [22].
  - Actuator dynamics and non-linear thruster allocation mapping (with reverse thrust efficiency loss).

3.2.1 USV Kinematics

- **State Representation:**
  - **Inertial Position Vector:** $\boldsymbol{\eta}_v = [x_v, y_v, \psi]^T \in \mathbb{R}^3$, representing planar positions (x_v, y_v) and heading angle ψ in $\{\mathcal{I}\}$.
  - **Body-Fixed Velocity Vector:** $\boldsymbol{\nu}_v = [u, v, r]^T \in \mathbb{R}^3$, representing surge velocity u, sway velocity v, and yaw angular rate r in $\{\mathcal{B}_v\}$.
- **Kinematic Equations:**
  - Mappings from body velocities to inertial rate of change:

$$\dot{x}_v = u \cos\psi - v \sin\psi \tag{3.4}$$

$$\dot{y}_v = u \sin\psi + v \cos\psi \tag{3.5}$$

$$\dot{\psi} = r \tag{3.6}$$

- **Assumptions & Physical Rationale:**
  - Motion is assumed strictly planar on the water surface ($z_v = h_v$).
  - Out-of-plane degrees of freedom (heave z, roll ϕ, pitch θ) are neglected in the control-oriented model due to high catamaran hydrostatic stability and to maintain real-time computational efficiency in the NMPC optimizer (while fully preserved in simulation).

3.2.2 Control-Oriented Dynamic Model (Fossen 3-DOF)

- **Model Purpose:** Horizontal 3-DOF control-oriented dynamic model in $\{\mathcal{B}_v\}$ according to Fossen's standard marine vessel equations [22] for real-time trajectory optimization.
- **Equations of Motion:**
  - Formulated in body-fixed frame $\{\mathcal{B}_v\}$:

$$\dot{u} = \frac{X - d_u(u)u}{m} + vr \tag{3.7}$$

$$\dot{v} = \frac{Y - d_v(v)v}{m} - ur \tag{3.8}$$

$$\dot{r} = \frac{N - d_r(r)r}{I_{zz}} \tag{3.9}$$

  - **Inertial Parameters:** m is the vessel mass, and I_{zz} is the moment of inertia about vertical axis $\mathbf{z}_v$.
  - **Virtual Control Inputs:** X (surge force), Y (sway force), and N (yaw moment) applied at the center of gravity.
  - **Coriolis / Centrifugal Coupling:** $+vr$ and $-ur$ account for acceleration components arising from the rotating reference frame $\{\mathcal{B}_v\}$.
- **Non-linear Hydrodynamic Damping Model:**
  - Combines linear and quadratic drag terms.
  - **Surge Damping with Directional Asymmetry:** Accounts for the geometrical difference between bow and stern:

$$d_u(u) = \begin{cases} X_U + X_{UU}|u|, & \text{if } u \geq 0 \quad \text{(forward motion)} \\ X_U + X_{UU\_BWD}|u|, & \text{if } u < 0 \quad \text{(backward motion)} \end{cases} \tag{3.10}$$

  - **Sway Damping:**

$$d_v(v) = Y_V + Y_{VV}|v| \tag{3.11}$$

  - **Yaw Damping:**

$$d_r(r) = N_R + N_{RR}|r| \tag{3.12}$$

- **Parameters:** Linear coefficients (X_U, Y_V, N_R) and quadratic coefficients ($X_{UU}, X_{UU\_BWD}, Y_{VV}, N_{RR}$).

- **Assumptions & Physical Rationale:**
  - **Planar Motion:** Motion is assumed strictly planar on the water surface ($z_v = h_v$).
  - **Out-of-Plane Dynamics:** Heave (z), roll (ϕ), and pitch (θ) are neglected in the control model due to high catamaran hydrostatic stability, maintaining computational tractability for NMPC (while active in the Gazebo simulation).
  - **Added Mass Simplification:** Hydrodynamic added mass effects are absorbed into the effective vessel inertia parameters (m, I_{zz}), preserving a diagonal and constant mass matrix for fast real-time NMPC optimization.

3.2.3 Thruster Allocation Mapping

- **Propulsion Configuration:**
  - Dual steerable stern thrusters (azimuth pods).
  - **Geometric Layout:** Located at longitudinal moment arm x_{arm} behind center of gravity (CG), and lateral moment arms $\pm y_{arm}$ relative to centerline.
  - **Actuator Inputs:** Thrust magnitudes T_L, T_R and azimuth angles α_L, α_R for port (left) and starboard (right) thrusters.

- **Reverse Thrust Efficiency Asymmetry:**
  - Captures propeller efficiency loss during reverse rotation due to blade profile curvature:

$$T_{i,eff} = \begin{cases} T_i, & \text{if } T_i \geq 0 \\ 0.746 \cdot T_i, & \text{if } T_i < 0 \end{cases} \quad \text{for } i \in \{L, R\} \tag{3.13}$$

    where 0.746 is the reverse thrust degradation multiplier.

- **Allocation Mapping Equations:**
  - Transformation from physical thruster states $(T_{L,eff}, T_{R,eff}, \alpha_L, \alpha_R)$ to virtual body forces $[X, Y, N]^T$ at CG:

$$X = T_{L,eff} \cos\alpha_L + T_{R,eff} \cos\alpha_R \tag{3.14}$$

$$Y = T_{L,eff} \sin\alpha_L + T_{R,eff} \sin\alpha_R \tag{3.15}$$

$$N = x_{arm} \left(T_{L,eff} \sin\alpha_L + T_{R,eff} \sin\alpha_R\right) - y_{arm} \left(T_{L,eff} \cos\alpha_L - T_{R,eff} \cos\alpha_R\right) \tag{3.16}$$

- **Actuator Bounds & Constraints:**

- Physical thrust saturation $T_{\min} \leq T_i \leq T_{\max}$ and steering limits $|\alpha_i| \leq \alpha_{\max}$ define the feasible virtual control bounds $[X_{\min}, X_{\max}] \times [-Y_{\max}, Y_{\max}] \times [-N_{\max}, N_{\max}]$ enforced in the NMPC solver.

- **Implementation Details:**
  - Embedded directly in the low-level C++ control pipeline to map optimal wrench $[X, Y, N]^T$ from the NMPC into individual physical thruster setpoints.

3.3 x500 Quadrotor Model

3.3.1 UAV Kinematics

The UAV operates in 3D airspace. Following standard aircraft modeling conventions [23], the position vector of the UAV CG in the inertial frame $\{\mathcal{I}\}$ is denoted by $\mathbf{p}_a = [x_a, y_a, z_a]^T$. The translational velocity vector of the vehicle, also expressed in the inertial reference frame $\{\mathcal{I}\}$, is defined as $\mathbf{v}_a = [v_{x,a}, v_{y,a}, v_{z,a}]^T$. The kinematic relationship mapping the rate of change of the global position coordinates to the inertial velocity vector is directly given by:

$$\dot{x}_a = v_{x,a} \tag{3.17}$$

$$\dot{y}_a = v_{y,a} \tag{3.18}$$

$$\dot{z}_a = v_{z,a} \tag{3.19}$$

Unlike the USV, whose velocity coordinates are defined in the body-fixed frame $\{\mathcal{B}_v\}$ to simplify hydrodynamic force mapping, the translational velocities of the quadrotor are formulated in the inertial frame $\{\mathcal{I}\}$. This representation aligns with the design of the control-oriented model and avoids unnecessary coordinate rotations in the translational dynamics. The orientation of the quadrotor body-fixed frame $\{\mathcal{B}_a\}$ relative to the inertial frame $\{\mathcal{I}\}$ is defined by the rotation matrix $\mathbf{R}_a(\phi, \theta, \psi_a)$ given in (3.2).

3.3.2 Control-Oriented Model (Attitude Lag)

To achieve a control-oriented model suitable for real-time execution in the NMPC framework, the full 6-DOF rigid-body dynamics of the quadrotor are simplified. Instead of modeling the high-frequency rotational dynamics, rotor speeds, and torque mixing, a cascaded control assumption is adopted. The high-rate attitude inner loop of the PX4 autopilot is assumed to track commanded roll and pitch references with a characteristic first-order response lag.

The actual roll ϕ and pitch θ dynamics are therefore modeled as:

$$\dot{\phi} = \frac{\phi_{cmd} - \phi}{\tau} \tag{3.20}$$

$$\dot{\theta} = \frac{\theta_{cmd} - \theta}{\tau} \tag{3.21}$$

where τ is the inner-loop attitude response time constant, and ϕ_{cmd} and θ_{cmd} represent the commanded roll and pitch angle setpoints, respectively.

To prevent aggressive variations in the commanded setpoints and ensure smooth trajectory planning, the state vector is augmented to include the commanded attitude variables. By defining the time derivatives of these commands as the virtual control inputs, we have:

$$\dot{\phi}_{cmd} = \dot{\phi}_{cmd,ctrl} \tag{3.22}$$

$$\dot{\theta}_{cmd} = \dot{\theta}_{cmd,ctrl} \tag{3.23}$$

where $\dot{\phi}_{cmd,ctrl}$ and $\dot{\theta}_{cmd,ctrl}$ are the control variables computed by the NMPC. This formulation allows the optimizer to directly penalize the rate of command changes in the cost function, enforcing actuator smoothness and preventing abrupt drone maneuvers that could destabilize the coupled tethered system.

3.3.3 Thrust Vectoring and Acceleration Kinematics

The translational motion of the UAV is driven by the collective thrust generated by the four rotors, which acts along the body-fixed vertical axis $\mathbf{z}_a$. The thrust force per unit mass (mass-normalized thrust acceleration) is denoted by a_T. Using the rotation matrix $\mathbf{R}_a(\phi, \theta, \psi_a)$ from the body frame $\{\mathcal{B}_a\}$ to the inertial frame $\{\mathcal{I}\}$ defined in (3.2), the thrust acceleration vector in the global East-North-Up (ENU) coordinate system is:

$$\mathbf{a}_{thrust} = \mathbf{R}_a(\phi, \theta, \psi_a) \begin{bmatrix} 0 \\ 0 \\ a_T \end{bmatrix} \tag{3.24}$$

By computing this product using the third column of the rotation matrix, the components of $\mathbf{a}_{thrust} = [a_{x,thrust}, a_{y,thrust}, a_{z,thrust}]^T$ are:

$$a_{x,thrust} = a_T\left(\cos\psi_a \sin\theta \cos\phi + \sin\psi_a \sin\phi\right) \tag{3.25}$$

$$a_{y,thrust} = a_T\left(\sin\psi_a \sin\theta \cos\phi - \cos\psi_a \sin\phi\right) \tag{3.26}$$

$$a_{z,thrust} = a_T \cos\theta \cos\phi \tag{3.27}$$

where ψ_a is the yaw heading angle of the drone, which is treated as a slowly varying external parameter in the control horizon.

Combining the thrust acceleration with the gravity vector $\mathbf{g} = [0, 0, -g]^T$, where $g = 9.81$ m/s^2 is the acceleration due to gravity, the complete translational dynamics of the quadrotor are formulated in the inertial frame $\{\mathcal{I}\}$ as:

$$\dot{v}_{x,a} = a_T\left(\cos\psi_a \sin\theta \cos\phi + \sin\psi_a \sin\phi\right) \tag{3.28}$$

$$\dot{v}_{y,a} = a_T\left(\sin\psi_a \sin\theta \cos\phi - \cos\psi_a \sin\phi\right) \tag{3.29}$$

$$\dot{v}_{z,a} = a_T \cos\theta \cos\phi - g \tag{3.30}$$

This formulation couples the attitude states (ϕ, θ) directly to the translational acceleration of the vehicle, allowing the NMPC to manipulate the quadrotor's trajectory in the 3D space by adjusting the attitude commands and thrust magnitude.

3.4 Tether and Winch Model

3.4.1 Control-Oriented Geometric Constraints

For the formulation of the real-time NMPC algorithm, modeling the full multi-body or lumped-mass catenary dynamics of the tether inside the optimization prediction horizon is computationally prohibitive. Instead, the coupling effect is modeled as a geometric boundary constraint. The tether is assumed to act as a spherical workspace envelope of radius L_{tether} centered at the USV anchor point $\mathbf{p}_{anchor} = [x_{boat}, y_{boat}, z_{boat}]^T$.

Assuming the USV moves on the water surface ($z_{boat} = 0$), the geometric distance d_{tether} between the UAV position $\mathbf{p}_a = [x_a, y_a, z_a]^T$ and the shipboard anchor is expressed as:

$$d_{tether} = \sqrt{(x_a - x_{boat})^2 + (y_a - y_{boat})^2 + z_a^2} \tag{3.31}$$

To enforce that the UAV remains within the physical boundary of the tether, a non-linear path constraint is formulated in the optimal control problem:

$$0 \leq h(\mathbf{x}, \mathbf{p}) \leq L_{tether}^2 \tag{3.32}$$

where the constraint expression $h(\mathbf{x}, \mathbf{p})$ is defined as the squared distance:

$$h(\mathbf{x}, \mathbf{p}) = (x_a - x_{boat})^2 + (y_a - y_{boat})^2 + z_a^2 \tag{3.33}$$

In the C++ control pipeline, the current boat coordinates $[x_{boat}, y_{boat}]^T$ are supplied to the solver as time-varying parameters $\mathbf{p}$ along the prediction horizon. To handle situations where external environmental perturbations (such as wind gusts) temporarily pull the quadrotor beyond the nominal tether length, this inequality is implemented as a soft constraint using quadratic slack variables. This prevents optimization solver infeasibility while maintaining a heavy penalty on constraint violations.

3.4.2 Virtual Winch Kinematics and Spooling Modes

The active winch system on the WAM-V deck dynamically manages the deployed tether length, denoted by L_{tether}. The rate of spooling $\dot{L}_{tether}$ is controlled to adapt the tether configuration to the cooperative mission requirements. Two operational modes are formulated for the virtual winch:

- **Geometric Spooling Mode:** The winch tracks the geometric distance d_{tether} between the vehicles. To prevent the cable from becoming overly taut and generating high disturbance forces on the UAV, a positive slack factor σ_{slack} is introduced. The target tether length is computed as:

$$L_{tether,cmd} = d_{tether} \cdot (1 + \sigma_{slack}) \tag{3.34}$$

where σ_{slack} represents a safety margin allowing a controlled catenary sag.

- **Tension-Regulated Spooling Mode:** The winch adjusts the spooling velocity to maintain a constant target tension T_{ref}. The control law for the winch spooling speed $\dot{L}_{tether}$ is defined as a function of the tension tracking error:

$$\dot{L}_{tether} = K_{T,p}\left(T_{meas} - T_{ref}\right) + K_{T,d}\dot{T}_{meas} \tag{3.35}$$

where T_{meas} is the measured tension at the anchor, and $K_{T,p}$, $K_{T,d}$ are the proportional and derivative gains of the tension controller, respectively.

3.4.3 Tether Coupling Force Model

To evaluate the loading effects on both the USV and the UAV without solving complex partial differential equations online, the physical force exerted by the tether on the vehicles is modeled using a simplified static representation. The tether force vector $\mathbf{F}_{tether}$ acting on the quadrotor is assumed to pull the vehicle directly toward the shipboard anchor point. The magnitude of this force is simplified to the static weight of the deployed (outside) cable segment plus a nominal base tension:

$$T_0 = T_{base} + \rho_L L_{tether} g \tag{3.36}$$

where ρ_L is the linear mass density of the cable (set to 0.020 kg/m in the simulation), L_{tether} is the length of the deployed segment outside the winch, and T_{base} is a constant representing the minimum pre-tension.

Assuming the line is straight and taut between the vehicles, the tether force vector $\mathbf{F}_{tether}$ experienced by the UAV is modeled as:

$$\mathbf{F}_{tether} = T_0 \frac{\mathbf{p}_{anchor} - \mathbf{p}_a}{\sqrt{\|\mathbf{p}_{anchor} - \mathbf{p}_a\|^2 + \epsilon^2}} \tag{3.37}$$

where $\mathbf{p}_a$ is the UAV position, $\mathbf{p}_{anchor}$ is the global anchor position on the ship deck, and ϵ is a small positive regularization constant (10^{-3}) added to prevent division by zero when the vehicles are coincident.

Chapter 4

Cooperative Control Design

This chapter presents the design and formulation of the cooperative control architecture for the coupled UAV-USV system. To achieve coordinated path following while ensuring tether tension stability and workspace safety, a decentralized control strategy is proposed. The control system is divided into three main components: a planar path-following NMPC for the WAM-V catamaran, a three-dimensional tracking NMPC for the x500 quadrotor, and an active spooling winch controller. The optimization formulations, cost functions, constraints, and real-time execution parameters of each controller are described in detail.

4.1 Cooperative Control Architecture Overview

The cooperative control strategy is structured in a decentralized leader-follower configuration, as illustrated in the block diagram of Figure 4.1. This architecture separates the control loops of each vehicle, allowing them to operate at different frequencies suited to their respective physical dynamics:

- **USV (Leader):** Operates at a frequency of $10\,$Hz ($\Delta t = 0.1\,$s). The USV NMPC tracks a planar reference path $\eta_{v,ref}$ on the water surface and outputs virtual forces and moment $[X, Y, N]^T$. These virtual controls are mapped by the thruster allocation block to steerable thruster forces and angles (T_i, α_i) to command the catamaran.

- **UAV (Follower):** Operates at a frequency of $50\,$Hz ($\Delta t = 0.02\,$s). The quadrotor tracks a 3D spatial trajectory $\mathbf{p}_{a,ref}$. To prevent tether overstretching and cable breaking, the UAV NMPC subscribes to the USV's global position and injects it as an external time-varying parameter vector $\mathbf{p}_{anchor}$ into the solver, dynamically constraining the UAV's prediction workspace. The controller outputs attitude command rates $(\dot{\phi}_{cmd}, \dot{\theta}_{cmd})$ and collective thrust acceleration a_T, which are cinematic-mapped to desired quaternions $\mathbf{q}_d$ and sent to the PX4 autopilot.

- **Active Winch:** Operates independently, monitoring the tether tension and Euclidean distance between vehicles to adjust the nominal deployed tether length L_{tether}.

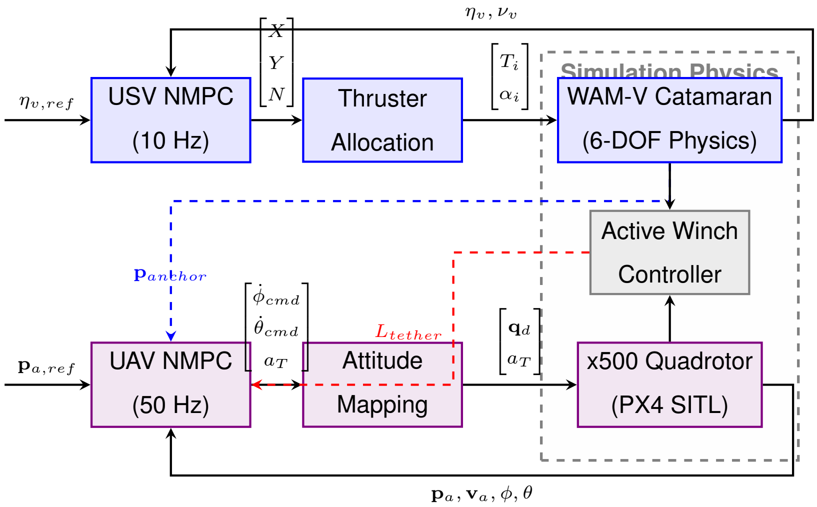


Figure 4.1: Control system architecture block diagram displaying the coupling, feedbacks, and communication flow between the USV, UAV, and active winch loop.

4.2 Nonlinear Model Predictive Control Principles

For a general non-linear continuous-time system governed by the state differential equations:

$$\dot{\mathbf{x}}(t) = \mathbf{f}\left(\mathbf{x}(t), \mathbf{u}(t), \mathbf{p}(t)\right) \tag{4.1}$$

where $\mathbf{x} \in \mathbb{R}^{n_x}$ is the state vector, $\mathbf{u} \in \mathbb{R}^{n_u}$ is the control input vector, and $\mathbf{p} \in \mathbb{R}^{n_p}$ is the parameter vector, the NMPC solves a discrete-time optimal control problem (OCP) at each sampling instance. The optimization problem is defined over a prediction horizon of N intervals, with a control step size T_s, solving:

$$\min_{\mathbf{x}_k, \mathbf{u}_k} \quad \sum_{k=0}^{N-1} \|\mathbf{h}(\mathbf{x}_k, \mathbf{u}_k, \mathbf{p}_k) - \mathbf{y}_{ref,k}\|_{\mathbf{W}}^2 + \|\mathbf{h}_e(\mathbf{x}_N, \mathbf{p}_N) - \mathbf{y}_{ref,N}\|_{\mathbf{W}_e}^2 \tag{4.2}$$

$$\text{s.t.} \quad \mathbf{x}_0 = \hat{\mathbf{x}}(t_0) \tag{4.3}$$

$$\mathbf{x}_{k+1} = \mathbf{\Phi}\left(\mathbf{x}_k, \mathbf{u}_k, \mathbf{p}_k\right), \quad k = 0, \ldots, N-1 \tag{4.4}$$

$$\mathbf{u}_{min} \leq \mathbf{u}_k \leq \mathbf{u}_{max}, \quad k = 0, \ldots, N-1 \tag{4.5}$$

$$\mathbf{x}_{min} \leq \mathbf{x}_k \leq \mathbf{x}_{max}, \quad k = 0, \ldots, N-1 \tag{4.6}$$

$$\mathbf{h}_{min} \leq \mathbf{h}(\mathbf{x}_k, \mathbf{u}_k, \mathbf{p}_k) \leq \mathbf{h}_{max}, \quad k = 0, \ldots, N-1 \tag{4.7}$$

where $\mathbf{h}(\mathbf{x}_k, \mathbf{u}_k, \mathbf{p}_k)$ is the stage cost output function, $\mathbf{h}_e(\mathbf{x}_N, \mathbf{p}_N)$ is the terminal cost output function, and $\mathbf{W}, \mathbf{W}_e$ are positive semi-definite weighting matrices of appropriate dimensions. The function $\mathbf{\Phi}$ represents the discrete-time system integration, which is solved using a fourth-order Runge-Kutta (ERK4) integration scheme. Equation (4.3) enforces the initial state constraint based on the current system state feedback $\hat{\mathbf{x}}(t_0)$.

To satisfy real-time constraints, the OCP is solved using the Real-Time Iteration (RTI) scheme within the Sequential Quadratic Programming (SQP) framework. The RTI scheme performs only a single SQP iteration per control interval, linearizing the dynamics and constraints around the previous solution trajectory. The resulting linearized Quadratic Programming (QP) subproblem is solved using the high-performance interior-point solver HPIPM, ensuring rapid execution.

4.3 UAV Guidance NMPC

The tracking NMPC for the x500 quadrotor is designed to command translation and attitude rates while satisfying workspace boundary constraints imposed by the tether length.

4.3.1 State, Control, and Parameter Vectors

The UAV guidance model is formulated with a 10-dimensional state vector $\mathbf{x}_a$, a 3-dimensional control input vector $\mathbf{u}_a$, and a 3-dimensional time-varying parameter vector $\mathbf{p}_a$:

$$\mathbf{x}_a = \begin{bmatrix} p_x & p_y & p_z & v_x & v_y & v_z & \phi & \theta & \phi_{cmd} & \theta_{cmd} \end{bmatrix}^T \tag{4.8}$$

$$\mathbf{u}_a = \begin{bmatrix} \dot{\phi}_{cmd,ctrl} & \dot{\theta}_{cmd,ctrl} & a_T \end{bmatrix}^T \tag{4.9}$$

$$\mathbf{p}_a = \begin{bmatrix} \psi_a & x_{boat} & y_{boat} \end{bmatrix}^T \tag{4.10}$$

where p_x, p_y, p_z are the 3D position coordinates and v_x, v_y, v_z are the translational velocity coordinates in the inertial reference frame $\{\mathcal{I}\}$. The angles ϕ, θ represent the actual roll and pitch of the vehicle, while ϕ_{cmd}, θ_{cmd} represent the commanded angles tracked by the autopilot inner loop. The control commands computed by the NMPC are the commanded roll and pitch rates $\dot{\phi}_{cmd,ctrl}, \dot{\theta}_{cmd,ctrl}$ along with the collective acceleration thrust magnitude a_T. The parameter vector captures the yaw heading ψ_a and the USV deck coordinates x_{boat}, y_{boat}.

4.3.2 OCP Cost Function and Penalties

The tracking objective is to minimize the deviation of the states and controls from a reference trajectory. The stage cost output vector $\mathbf{h}_a$ and the terminal cost output vector $\mathbf{h}_{a,e}$ are defined as:

$$\mathbf{h}_a(\mathbf{x}_a, \mathbf{u}_a, \mathbf{p}_a) = \begin{bmatrix} \mathbf{x}_a^T & \mathbf{u}_a^T \end{bmatrix}^T \tag{4.11}$$

$$\mathbf{h}_{a,e}(\mathbf{x}_a, \mathbf{p}_a) = \mathbf{x}_a \tag{4.12}$$

The weighting matrices $\mathbf{W}_a$ and $\mathbf{W}_{a,e}$ are configured diagonally to scale individual penalties:

$$\mathbf{W}_a = \text{diag}\left(\mathbf{Q}_{pos}, \mathbf{Q}_{vel}, \mathbf{Q}_{tilt}, \mathbf{Q}_{tilt\_cmd}, \mathbf{R}_{rates}, R_{thrust}\right) \tag{4.13}$$

$$\mathbf{W}_{a,e} = \text{diag}\left(\mathbf{Q}_{pos}, \mathbf{Q}_{vel}, \mathbf{Q}_{tilt}, \mathbf{Q}_{tilt\_cmd}\right) \tag{4.14}$$

where the positional penalty $\mathbf{Q}_{pos} = \text{diag}(20, 20, 20)$ is heavily weighted to enforce path tracking, and $\mathbf{R}_{rates} = \text{diag}(15, 15)$ penalizes large attitude command rates to guarantee smooth flight.

4.3.3 Actuator and Workspace Constraints

The bounds on states and control inputs represent the physical flight envelope of the x500 quadrotor:

$$-\mathbf{v}_{max} \leq \begin{bmatrix} v_x & v_y & v_z \end{bmatrix}^T \leq \mathbf{v}_{max} \tag{4.15}$$

$$-\phi_{max} \leq \phi, \theta, \phi_{cmd}, \theta_{cmd} \leq \phi_{max} \tag{4.16}$$

$$-\dot{\phi}_{max} \leq \dot{\phi}_{cmd,ctrl}, \dot{\theta}_{cmd,ctrl} \leq \dot{\phi}_{max} \tag{4.17}$$

$$a_{T,min} \leq a_T \leq a_{T,max} \tag{4.18}$$

with $v_{max} = 15.0$ m/s, $\phi_{max} = 0.4$ rad ($\approx 23°$), $\dot{\phi}_{max} = 0.8$ rad/s, $a_{T,min} = 0.1 \cdot g$, and $a_{T,max} = 1.6 \cdot g$.

The tether constraint is implemented as a slacked nonlinear path inequality:

$$0 \leq (p_x - x_{boat})^2 + (p_y - y_{boat})^2 + p_z^2 \leq L_{tether}^2 + s \tag{4.19}$$

where $s \geq 0$ is a scalar slack variable. The cost function is augmented with linear and quadratic penalties on s, scaled by weights z_l, Z_l, to strictly penalize tether overstretch:

$$J_{slack} = z_l s + Z_l s^2 \tag{4.20}$$

with $z_l = 500$ and $Z_l = 10000$, ensuring that the tether boundary is maintained as a soft but highly protected constraint.

4.3.4 Attitude Command Mapping (PX4 Interface)

The NMPC outputs attitude setpoints ϕ_{cmd} and θ_{cmd} along with thrust a_T at each interval. To command the PX4 flight controller, these outputs must be converted to a desired quaternion setpoint $\mathbf{q}_d = [q_w, q_x, q_y, q_z]^T$ in the local NED/FRD frame. The desired rotation matrix in the global ENU frame is first computed from the roll and pitch commands and the current yaw parameter:

$$\mathbf{R}_{enu} = \mathbf{R}_z(\psi_a)\mathbf{R}_y(\theta_{cmd})\mathbf{R}_x(\phi_{cmd}) \tag{4.21}$$

This rotation matrix is transformed into the vehicle's local frame using:

$$\mathbf{R}_d = \mathbf{M}_{world}\mathbf{R}_{enu}\mathbf{M}_{body} \tag{4.22}$$

where $\mathbf{M}_{world}$ and $\mathbf{M}_{body}$ are coordinate transformation matrices. The resulting rotation matrix $\mathbf{R}_d$ is converted to the desired Hamiltonian quaternion $\mathbf{q}_d$, which is published at high frequency to PX4's offboard attitude setpoint subscriber.

4.4 USV Tracking NMPC

The control objective for the USV NMPC is to guide the WAM-V catamaran along a planar path while managing the large inertia and hydrodynamic damping of the hull.

4.4.1 State and Control Vectors

The USV dynamic NMPC uses a 6-dimensional state vector $\mathbf{x}_v$ and a 3-dimensional control input vector $\mathbf{u}_v$:

$$\mathbf{x}_v = \begin{bmatrix} x_{pos} & y_{pos} & \psi & u & v & r \end{bmatrix}^T \tag{4.23}$$

$$\mathbf{u}_v = \begin{bmatrix} X_{force} & Y_{force} & N_{moment} \end{bmatrix}^T \tag{4.24}$$

where x_{pos}, y_{pos} are the planar position coordinates, ψ is the heading yaw angle, and u, v, r are the surge, sway, and yaw rate velocities in the body-fixed frame $\{\mathcal{B}_v\}$. The control vector consists of the virtual surge force X_{force}, sway force Y_{force}, and yaw moment N_{moment} applied at the catamaran's center of gravity.

4.4.2 OCP Cost Function and Penalties

The tracking cost function minimizes the tracking errors of states and regulates virtual inputs. The stage weighting matrix $\mathbf{W}_v$ is defined as:

$$\mathbf{W}_v = \text{diag}\left(Q_x, Q_y, Q_\psi, Q_u, Q_v, Q_r, R_x, R_y, R_n\right) \tag{4.25}$$

To eliminate position lag and keep the vessel heading tangent to the trajectory, the position weights are set to $Q_x = Q_y = Q_\psi = 150.0$. The control regularization weights are set to small values ($R_x = 10^{-4}, R_y = 10^{-3}, R_n = 5 \cdot 10^{-4}$) to allow the thrusters to generate sufficient force to overcome hydrodynamic drag.

4.4.3 Dynamic Actuation Limits

The bounds on the virtual controls represent the maximum physical limits of the WAM-V steerable stern thrusters, mapped to the center of gravity:

$$X_{min} \leq X_{force} \leq X_{max} \tag{4.26}$$

$$-Y_{max} \leq Y_{force} \leq Y_{max} \tag{4.27}$$

$$-N_{max} \leq N_{moment} \leq N_{max} \tag{4.28}$$

where $X_{max} = 1020\,$N (maximum forward thrust), $X_{min} = -760\,$N (maximum reverse thrust accounting for the 0.746 efficiency factor), $Y_{max} = 510\,$N (lateral force limit for roll stability), and $N_{max} = 1200\,$N $\cdot$ m (maximum steering moment).

4.5 Active Winch Control Strategy

The active winch system acts as a dynamic coordinator, adjusting the deployed cable length L_{tether} based on the flight phase to manage tether tension and slack. It operates in two main modes:

- **Geometric Spooling Mode (Flight/Cruise Phase):** During autonomous patrol and pursuit phases, the winch maintains a controlled slack using a safety margin factor σ_{slack} (e.g., 0.1 to 0.2) relative to the instantaneous distance between the vehicles:

$$L_{tether,cmd} = d_{tether} \cdot (1 + \sigma_{slack}) \tag{4.29}$$

  This controlled slack is essential in rough seas: by allowing the cable to hang in a slack catenary curve, the dynamic heave, pitch, and roll motions of the catamaran under wave action are mechanically decoupled from the UAV, preventing tension disturbances from destabilizing the drone's flight.

- **Tension-Guided Landing (TGL) Mode (Landing Phase):** During landing and recovery, the winch transitions to a high-tension regulation mode. A proportional-derivative (PD) controller regulates the spooling velocity $\dot{L}_{tether}$ to maintain a high target tension $T_{landing}$ (e.g., 15 N to 20 N), pulling the cable taut:

$$\dot{L}_{tether} = K_{T,p} \left(T_{meas} - T_{landing}\right) + K_{T,d}\dot{T}_{meas} \tag{4.30}$$

  where $K_{T,p}$ and $K_{T,d}$ are the tracking gains, and T_{meas} is the measured tension. This taut cable acts as a physical guideway, pulling the UAV toward the center of the deck and preventing lateral drift from wind gusts or catamaran roll, enabling safe docking on the moving vessel without requiring predictive heave synchronization.

Chapter 5

Simulation Environment and Software Implementation

This chapter describes the co-simulation setup, software environment, and virtual platforms used to implement and validate the cooperative control architecture. The system is simulated with high physical fidelity. The software stack integrates the ROS 2 workspace, the PX4 Autopilot SITL firmware, the MoorDyn cable physics engine, and the Gazebo simulator. The following sections detail the software architecture, the virtual vehicle platforms, the tether and winch models, and the system integration and data flow.

5.1 Software Architecture and Workspace Structure

The system software is organized into modular ROS 2 packages running on top of the Linux Ubuntu 22.04 LTS operating system and the ROS 2 Humble Hawksbill middleware. This architecture ensures decoupling between control, estimation, simulation, and mission coordination.

5.1.1 Workspace Structure and Package Modularity

The ROS 2 workspace is organized into five primary packages, summarized in Table 5.1.

Table 5.1: Summary of the modular ROS 2 packages in the cooperative workspace.

Package Name	Language	Primary Responsibility
`bringup`	Python/YAML	System launch files, configuration files, and coordinate frame parameters.
`uav_mpc`	C++/Python	UAV Acados solver generation, telemetry mapping, and position NMPC.
`usv_mpc`	C++/Python	USV Acados solver generation, thruster allocation mapping, and tracking NMPC.
`moordyn_tether`	C++	MoorDyn mooring line node wrapper and active winch controller.
`ats_mission`	C++	Mission coordinator managing takeoff, orbiting, landing, and FSM transitions.

5.1.2 Code Design Pattern: Library and ROS Wrapper

To ensure code standard compliance, portability, and readability, all control nodes are implemented using a strict three-layer design pattern:

1. **The Entry Point (`_main.cpp`):** A minimal file containing only the main initialization function, instantiating the node wrapper, and spinning the ROS 2 executor.
2. **The ROS Wrapper (`_node.cpp/hpp`):** Handles all ROS-related tasks. It loads configuration parameters from YAML files, initializes publishers and subscribers, sets up timers, and manages the publishing of tf2 coordinate frame transformations. No mathematical calculations or control-oriented logic are written in this layer.
3. **The Logic Pipeline (`_pipeline.cpp/hpp`):** A pure C++ class that is completely independent of ROS. It uses native C++ and Eigen library types to execute the calculations. It wraps the Acados C structure, updates the states and parameters, runs the optimization solver, and maps virtual forces to actuator commands.

To support multi-robot simulation and prevent topic collisions, all topic subscriptions and publications inside the nodes are written using relative topic names. This allows namespaces (e.g., `/wamv1/odom` and `/wamv2/odom`) to isolate vehicle telemetry.

5.1.3 Software Environment and Code Generation

For the design and generation of the optimization solvers of the NMPC controllers, a symbolic code generation pipeline is employed:

- **CasADi Symbolic Framework:** A Python script is written for each vehicle (UAV and USV) to define the state, control, and parameter variables symbolically. CasADi handles the automatic differentiation, compiling the non-linear dynamics equations and evaluating analytical Jacobians and Hessian approximations.

- **Acados Template Library:** Consumes the CasADi symbolic models to generate highly optimized, native C code solvers. Acados bypasses generic optimization libraries by generating customized solvers tailored to the specific OCP structure.

- **High-Performance Kernels:** The exported C code is compiled using the BLASFEO (Basic Linear Algebra Subprograms for Embedded Optimization) library, which optimizes processor cache utilization for small-to-medium-sized matrices. The linearized QP subproblems are solved using HPIPM (High-Performance Interior-Point Method), which implements a structured primal-dual interior-point method.

The temporal configuration of the NMPC solvers is configured independently for the two platforms to balance physical response rates and solver execution times:

- **UAV Solver:** Discretized with a control period of $T_{s,UAV} = 0.02$ s (50 Hz) and a prediction horizon of $N_{UAV} = 50$ intervals, yielding a total prediction horizon of $T_{f,UAV} = 1.0$ s.

- **USV Solver:** Discretized with a control period of $T_{s,USV} = 0.1$ s (10 Hz) and a prediction horizon of $N_{USV} = 20$ intervals, yielding a total prediction horizon of $T_{f,USV} = 2.0$ s.

5.2 Virtual Vehicle Platforms

The cooperative multi-agent system comprises two virtual vehicles: the WAM-V surface vessel and the Holybro x500 quadrotor.

5.2.1 WAM-V Catamaran (USV)

The Wave Adaptive Modular Vessel (WAM-V) is a unique catamaran consisting of two inflatable hulls connected to a central deck structure. Unlike rigid-hulled vessels, the WAM-V uses an articulating suspension system that isolates the payload deck from wave-induced pitch and roll to keep the deck stable. Sourced from the Virtual RobotX (VRX) simulation framework [24], the virtual platform matches the physical parameters of the real WAM-V used in international maritime robotics competitions. The virtual catamaran representation in Gazebo is shown in Figure 5.1.

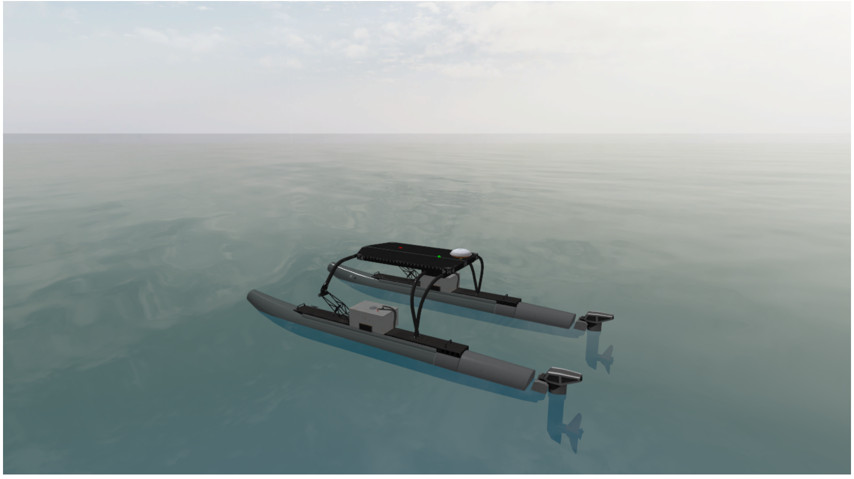
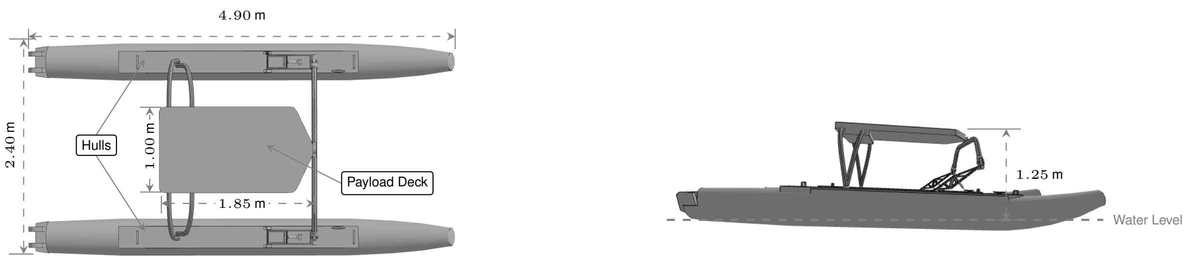


Figure 5.1: Virtual representation of the Wave Adaptive Modular Vessel (WAM-V) catamaran in the simulated environment.

Structurally, the 16-foot catamaran has an overall length of 4.90 m and an overall beam of 2.40 m. The central payload deck, measuring 1.85 m in length and 1.00 m in width, is supported by two parallel hulls at about 1.25 m above the water surface, as illustrated in Figure 5.2. In the simulator, the boat's total mass is modeled at 259.10 kg, which accounts for the catamaran chassis, the onboard batteries, the dual propulsion steering pods, and the sensory payloads.



(a) Top view.

(b) Side view.

Figure 5.2: WAM-V catamaran layout.

Instead of using a traditional rudder, the WAM-V steers by rotating its two rear propulsion pods, known as azimuthing thrusters, which are positioned at the stern of each hull, as shown in Figure 5.1. The control system developed in this work restricts their steering angle to a physical limit of ± 30 degrees to avoid twisting the power cables, with the steering rate limited to 1.0 rad/s.

The thrust force is adjusted by changing the propeller speed. The maximum thrust limit of each motor is set to 510.0 N in the forward direction. This represents the physical capabilities of commercial electric outboards commonly used on the WAM-V 16 in the RobotX competition, such as the Torqeedo Cruise 2.0 [25]. Additionally, the motors exhibit reduced efficiency in reverse due to propeller blade geometry; this is modeled with a reverse thrust efficiency multiplier of 0.746 that limits the maximum reverse thrust to -380.0 N.

Buoyancy is simulated by continuously calculating the volume of the hulls submerged under the water surface. The sea surface is modeled using Gerstner waves, and the simulator checks the hull geometry against the wave profile to resolve the hydrostatic forces. This yields natural lifting forces and moments that pitch and roll the catamaran in response to wave excitation. Hydrodynamic drag is represented through a combination of linear damping coefficients for low-speed viscous resistance and quadratic coefficients to capture high-speed form drag. These drag forces and moments are applied across all six degrees of freedom, as detailed in Table 5.2, so that the catamaran decelerates naturally when propulsion is cut off.

Table 5.2: Hydrodynamic damping coefficients of the virtual WAM-V catamaran.

Damping Mode	**Linear Coefficient**	**Quadratic Coefficient**	**Unit**
Surge	100.00	150.00	N$/$(m/s) or N$/$(m/s)^2
Sway	100.00	100.00	N$/$(m/s) or N$/$(m/s)^2
Heave	500.00	–	N$/$(m/s)
Roll	300.00	600.00	N $\cdot$ m$/$(rad/s) or N $\cdot$ m$/$(rad/s)^2
Pitch	900.00	900.00	N $\cdot$ m$/$(rad/s) or N $\cdot$ m$/$(rad/s)^2
Yaw	800.00	800.00	N $\cdot$ m$/$(rad/s) or N $\cdot$ m$/$(rad/s)^2

5.2.2 x500 Quadrotor (UAV)

The aerial platform is modeled after the Holybro x500 quadrotor frame. It is represented as a rigid body with 6 DOFs, consisting of a central hub and four rotating propeller links. The quadrotor has a total mass of 2.06 kg, consisting of a 2.00 kg central body and four rotor links of 0.016 kg each. The moments of inertia are $I_{xx,a} = 0.0217$ kg $\cdot$ m^2 for roll, $I_{yy,a} = 0.0217$ kg $\cdot$ m^2 for pitch, and $I_{zz,a} = 0.0400$ kg $\cdot$ m^2 for yaw. The propulsion system consists of four brushless motors in an X-configuration with 0.254 m (10-inch) propellers.

Rotor thrust and torque are calculated using a multicopter motor model based on propeller rotational speed. The motor constant is set to 8.55×10^{-6} N$/$(rad/s)^2 and the torque constant is 0.016 m. The maximum speed of the propellers is limited to 1000 rad/s. Motor responsiveness includes a speed lag modeled with time constants ($\tau_{up} = 0.0125$ s and $\tau_{down} = 0.0250$ s). The physical and aerodynamic parameters are summarized in Table 5.3.

Table 5.3: Physical and aerodynamic parameters of the virtual x500 quadrotor.

Parameter	**Symbol**	**Value**
Total Mass	m_a	2.064 kg
Base Link Mass	$m_{base,a}$	2.000 kg
Rotor Link Mass (each)	m_{rotor}	0.016 kg
Moment of Inertia (Roll)	$I_{xx,a}$	0.0217 kg $\cdot$ m^2
Moment of Inertia (Pitch)	$I_{yy,a}$	0.0217 kg $\cdot$ m^2
Moment of Inertia (Yaw)	$I_{zz,a}$	0.0400 kg $\cdot$ m^2
Motor Constant	K_{motor}	8.54858×10^{-6} N/(rad/s)^2
Moment Constant	K_{moment}	0.016 m
Max Rotational Velocity	Ω_{max}	1000.0 rad/s
Rotor Drag Coefficient	$C_{drag,r}$	8.06428×10^{-5}
Rolling Moment Coefficient	$C_{roll,r}$	1.00000×10^{-6}
Upward Motor Time Constant	τ_{up}	0.0125 s
Downward Motor Time Constant	τ_{down}	0.0250 s

These motor characteristics shape the quadrotor's dynamic response: speed lag limits rapid thrust variations, while drag coefficients account for propeller resistance.

5.3 Tether and Winch Integration

A flexible tether, managed by an active winch system on the catamaran's deck, physically couples the WAM-V catamaran and the x500 quadrotor.

5.3.1 Tether Representation Models

The tether can be simulated using a multibody joint chain or a lumped-mass model. The multibody joint chain is resolved in Gazebo. The lumped-mass model is resolved using the MoorDyn library [26].

The multibody joint chain is used for visual rendering and contact collisions. It consists of a chain of 30 rigid cylindrical links connected by spherical ball joints. Each link has a radius of 0.01 m and a mass of 0.02 kg. The total mass of the nominal 5.0 m chain is 0.60 kg. Joint damping is set to 2.0 N $\cdot$ s/m and friction to 0.1 N$\cdot$m. The first and last links are scaled to 20% of their length to prevent numerical instability at the attachment points.

The MoorDyn lumped-mass model is used for high-fidelity tension propagation. The cable is discretized into 30 mass nodes connected by spring-damper segments. The physical parameters represent an industrial micro-tether, with a diameter of 0.004 m, linear mass density of 0.020 kg/m, extensional stiffness of $EA = 100.0$ N, bending stiffness of $EI = 2.0 \times 10^{-5}$ N $\cdot$ m^2, and damping ratio of $\zeta = 0.8$. Drag coefficients are set to 1.2 transversely and 0.005 axially. The physical properties are summarized in Table 5.4.

Table 5.4: Physical and mechanical properties of the tether simulation models.

Property	**Symbol**	**Multibody Chain**	**MoorDyn Model**
Number of Segments	N_{seg}	30	30
Tether Diameter	D_{tether}	0.020 m	0.004 m
Linear Mass Density	ρ_{linear}	0.120 kg/m	0.020 kg/m
Total Mass (5.0 m cable)	m_{tether}	0.600 kg	0.100 kg
Extensional Stiffness	EA	–	100.0 N
Bending Stiffness	EI	–	2.0×10^{-5} N $\cdot$ m^2
Structural Damping Ratio	ζ	–	0.8
Joint Damping Coefficient	b_{joint}	2.0 N $\cdot$ s/m	–
Joint Friction Torque	τ_{fric}	0.1 N $\cdot$ m	–
Transverse Drag Coefficient	C_d	–	1.2
Axial Drag Coefficient	$C_{d,ax}$	–	0.005

The MoorDyn model represents a thinner, lighter cable that captures aerodynamic drag forces (Table 5.4).

5.3.2 Active Winch and Anchor Coupling

The active winch is mounted on the WAM-V deck, offset by $\mathbf{p}_{\text{anchor},v} = [-0.8, 0.0, 0.0]^T$ m from the center of gravity. The winch dynamically spools the tether to adjust its unstretched length L_{tether} between 1.5 m and 50.0 m, with a maximum spooling speed of 10.0 m/s.

The winch operates in two control modes. In geometric mode, the tether length is set to 115% of the line-of-sight distance between the vehicles, with the slack factor preventing sudden tension spikes. In tension mode, a PD controller regulates the spooling rate to maintain a target tension of $T_{target} = 0.1$ N, using a proportional gain of $K_p = 1.8$ m/(s $\cdot$ N) and a derivative gain of $K_d = 0.1$.

The physical coupling forces and winch spooling rates are simulated through the integration of the MoorDyn mooring line library in the ROS 2 node `moordyn_tether_node`. The C++ node tracks the relative transformation between the USV deck anchor point and the UAV center of gravity using the `tf2` listener. At a frequency of 100 Hz, the node updates the coordinates of the boundary nodes inside the MoorDyn solver with the tracked positions. Simultaneously, it listens to winch spooling commands from the winch controller (`moordyn_tether_winch`) and updates the unstretched line length property inside the MoorDyn solver. MoorDyn computes the internal segment tensions and propagates the resulting force vectors back to the boundary nodes; these are applied directly to the anchor links of the WAM-V and the x500 rigid bodies in Gazebo to couple the vehicles bidirectionally.

5.4 System Integration and Co-Simulation

To run and coordinate the simulation, a co-simulation environment is built in Gazebo Harmonic [27]. The software stack integrates the simulator with the control pipeline through communication bridges, as shown in Figure 5.3.

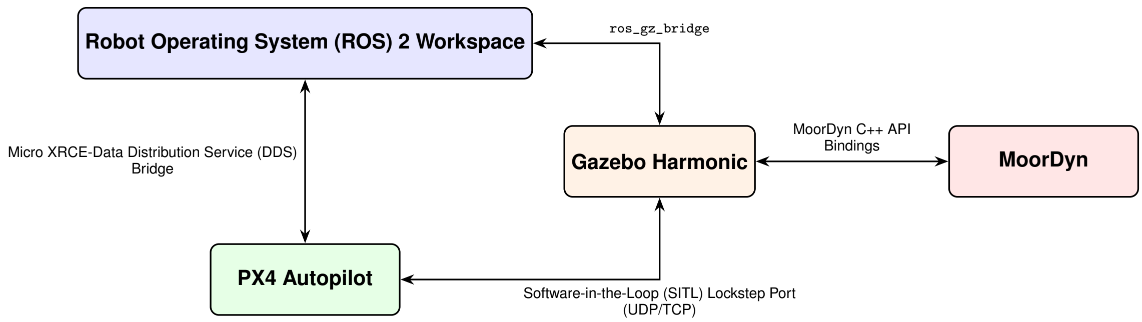


Figure 5.3: Simulation software ecosystem and communication interfaces.

5.4.1 PX4 SITL and micro XRCE-DDS Bridge

The x500 quadrotor simulation runs the actual PX4 Autopilot flight firmware in a Software-in-the-Loop (SITL) wrapper. The firmware runs on the host computer, receiving state feedback from Gazebo and commanding the virtual motor models via cascaded PID loops. The Micro XRCE-DDS bridge manages the high-rate, low-latency exchanges between the PX4 middleware and the ROS 2 network, allowing outer-loop NMPC algorithms to subscribe to state estimation (odometry) and publish offboard attitude and thrust targets.

5.4.2 ROS-Gazebo Bridge and Time Synchronization

The `ros_gz_bridge` maps the simulation clock, joint state commands (steerable thruster angles, propeller speeds), and winch actuator commands from ROS 2 to the Gazebo physics engine. To prevent time drift between the simulator and the controller nodes, all ROS 2 nodes are configured to run using simulation time. The Gazebo simulator publishes the simulation clock on the `/clock` topic, which is used by the ROS 2 executor to synchronize node timers.

5.4.3 Centralized Data Flow and Cooperative FSM

To coordinate the simulation, a centralized data flow architecture is implemented. The overall loop involves multiple distributed nodes communicating via ROS 2 topics and services. The ground-truth state feedback is published by the Gazebo and PX4 SITL instances. This telemetry is translated by the communication bridges and published to the ROS 2 network.

The control algorithms process this feedback to execute three main control loops:

- **USV Control Loop (10 Hz):** The Nonlinear Model Predictive Control (NMPC) node receives the vessel's ground-truth odometry and calculates the virtual forces and moment $[X, Y, N]^T$. The thruster allocation block maps these virtual forces to steering angles and motor thrust commands published to Gazebo.

- **UAV Control Loop (50 Hz):** The NMPC node receives the drone's ground-truth telemetry and the relative USV position to calculate collective thrust and attitude rates. These targets are published to the PX4 flight controller via Micro XRCE-DDS.
- **Tether and Winch Loop (50 Hz):** The MoorDyn node receives vehicle poses from TF, updates the cable state, and publishes the calculated tension forces to Gazebo. Simultaneously, the winch controller computes the spooling rate command $\dot{L}_{tether}$ to update the unstretched length of the cable in MoorDyn.

The mission coordinator node (`mission_manager`) in the `ats_mission` package orchestrates the flight phases of the drone relative to the ship using a Finite State Machine (FSM), as illustrated in Figure 5.4:

- **IDLE:** Initial state. Nodes check sensor telemetry, establish the tether connection frames, and wait for the start command.
- **TAKEOFF:** Triggers autonomous UAV takeoff. The quadrotor rises vertically to a hover altitude directly above the WAM-V helideck ($z_a = 3.0$ m relative to the deck). The USV and winch remain stationary.
- **COOPERATIVE_TRACKING:** The active phase. The USV starts tracking its planar trajectory, and the UAV starts its orbit tracking or path-following sequence centered on the USV anchor. The winch is activated in either geometric or tension-regulated spooling mode to manage tether slack.
- **LANDING:** Triggered at the end of the mission. The USV reduces speed or hovers. The UAV approaches the catamaran deck while the winch spools in the tether, maintaining line tension to guide the quadrotor safely onto the deck.

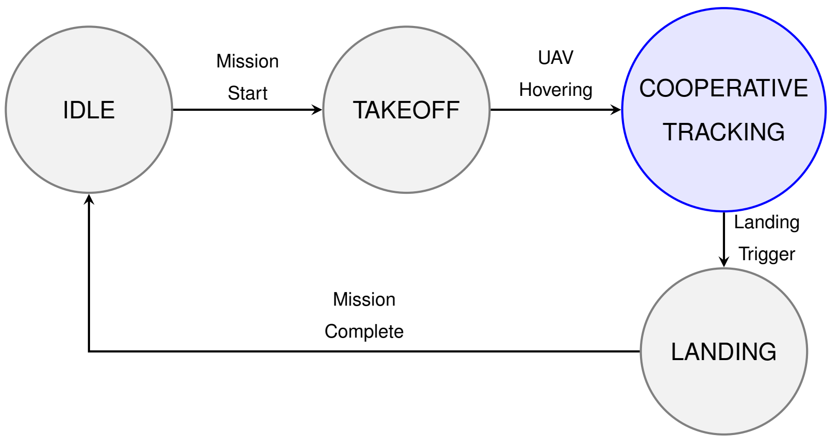


Figure 5.4: Finite State Machine (FSM) state transition diagram for the cooperative mission coordinator.

Chapter 6

Results and Discussion

This chapter presents the experimental identification, model validation, and closed-loop simulation results of the cooperative tethered UAV-USV system. First, the simplified control-oriented models derived in Chapter 3 are validated and calibrated against high-fidelity simulator data, and the UAV inner-loop attitude response lag is experimentally identified. Second, the individual and cooperative trajectory tracking precision of both vehicles is evaluated. Third, the enforcement of NMPC constraints and real-time solver execution times are analyzed. Finally, the complete cooperative control stack is validated under an integrated search and pursuit mission targeting a teleoperated target, as well as a comparative study evaluating tension-guided landing in rough seas.

6.1 Model Validation and Parameter Identification

Before evaluating the control performance, this section validates and calibrates the control-oriented kinematic and dynamic models derived in Chapter 3 using simulation telemetry and flight test data.

6.1.1 USV Model Identification and Calibration

To calibrate and validate the horizontal 3-DOF dynamic model of the WAM-V, a systematic two-step identification process is conducted. First, a set of dynamic excitation maneuvers is performed to collect estimation data. Second, the parameters are optimized by minimizing the difference between simulated and simulated-ground-truth states. Finally, the model is validated using a separate, independent excitation dataset to ensure generalization.

The parameter calibration relies on distinct excitation signals. The excitation maneuvers are designed to decouple the vehicle dynamics across all active degrees of freedom. Figure 6.1 shows the multi-phased excitation signals used for model validation. The estimation phase and validation phase use different signal profiles to prevent overfitting. The excitation profile consists of four primary stages: Phase 1 introduces pure surge steps to identify longitudinal damping, Phase 2 applies differential thrust commands to excite yaw rate, Phase 3 forces lateral motion to isolate sway damping, and Phase 4 provides fully coupled random actuation to capture cross-coupling dynamics.

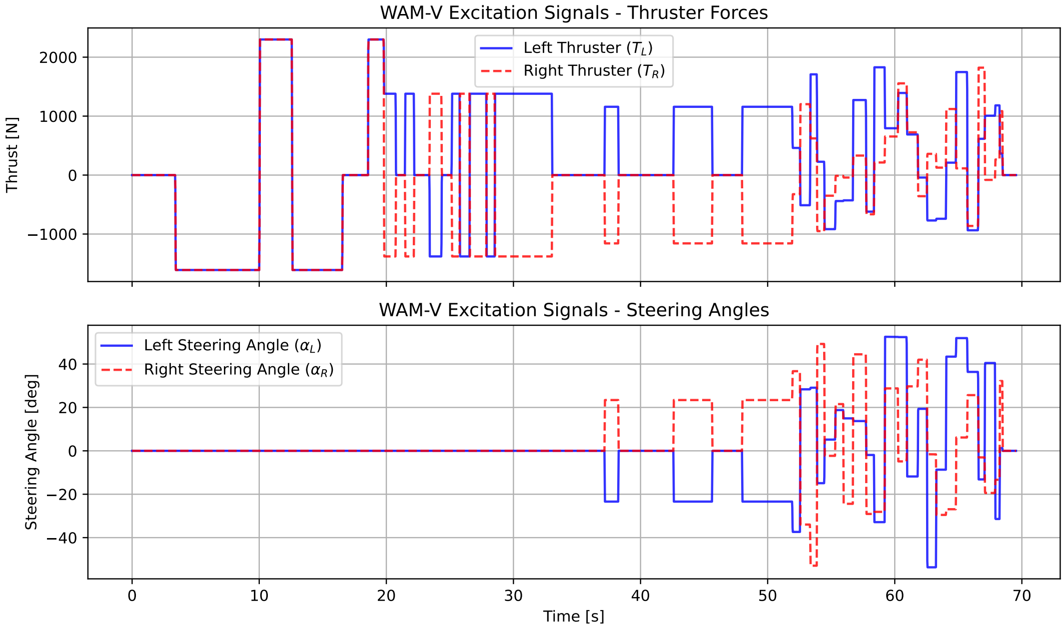


Figure 6.1: Multi-phased excitation signals command profiles (forces and angles) for the WAM-V parameter validation.

The dynamic parameters are calibrated using a grey-box estimation approach. The optimization process fits the Fossen 3-DOF model equations to the collected Gazebo simulator telemetry. This is achieved by solving a non-linear least-squares optimization problem. The cost function minimizes the tracking error of the linear and angular velocities:

$$\min_{\theta} \sum_{k=1}^{N} \|\nu_{v,actual}(k) - \nu_{v,pred}(k, \theta)\|_2^2 \tag{6.1}$$

where θ is the vector of unknown hydrodynamic parameters. The predicted velocity vector $\nu_{v,pred}$ is computed by integrating the dynamics using a fourth-order Runge-Kutta solver.

Table 6.1 presents the resulting identified parameters. It compares the optimized values against the nominal parameters specified in the Gazebo simulation descriptor file.

Table 6.1: Comparative summary of WAM-V physical and hydrodynamic parameters.

Parameter	Symbol	Identified (MPC)	Gazebo (Nominal)	Unit
Rigid-Body Mass	m	289.83	259.10	kg
Yaw Moment of Inertia	I_{zz}	600.00	527.70	kg · m^2
Linear Surge Damping	X_U	165.97	100.00	N/(m/s)
Quadratic Surge Damping (Bow)	X_{UU}	140.00	150.00	N/(m/s)^2
Quadratic Surge Damping (Stern)	X_{UU_BWD}	157.68	150.00	N/(m/s)^2
Linear Sway Damping	Y_V	92.85	100.00	N/(m/s)
Quadratic Sway Damping	Y_{VV}	102.42	100.00	N/(m/s)^2
Linear Yaw Damping	N_R	400.76	800.00	N · m/(rad/s)
Quadratic Yaw Damping	N_{RR}	1069.15	800.00	N · m/(rad/s)^2
Thruster Longitudinal Arm	x_{arm}	-2.37	-2.37	m
Thruster Lateral Arm Offset	y_{arm}	1.03	1.03	m

As shown in Table 6.1, the identified rigid-body mass m and yaw inertia I_{zz} are higher than the physical nominal values. This discrepancy is physically expected. The optimization process absorbs the hydrodynamic added mass effects. The simplified dynamic model does not explicitly account for these added mass coefficients.

To verify the predictive accuracy of the identified model, we compute the Root Mean Square Error (RMSE) and the FIT percentage metric. The FIT percentage metric is defined as:

$$FIT_{x_i} = 100 \cdot \left(1 - \frac{\sqrt{\sum_{k=1}^{N} \left(x_{i,actual}(k) - x_{i,pred}(k)\right)^2}}{\sqrt{\sum_{k=1}^{N} \left(x_{i,actual}(k) - \bar{x}_{i,actual}\right)^2}}\right) \tag{6.2}$$

where $x_{i,actual}$ and $x_{i,pred}$ are the actual and predicted states, respectively, and $\bar{x}_{i,actual}$ is the mean of the actual state.

Figure 6.2 shows the comparison of the predicted and actual velocities under the validation dataset. The velocity profiles display high agreement. This confirms that the linear and non-linear damping parameters generalize well to unseen maneuvers.

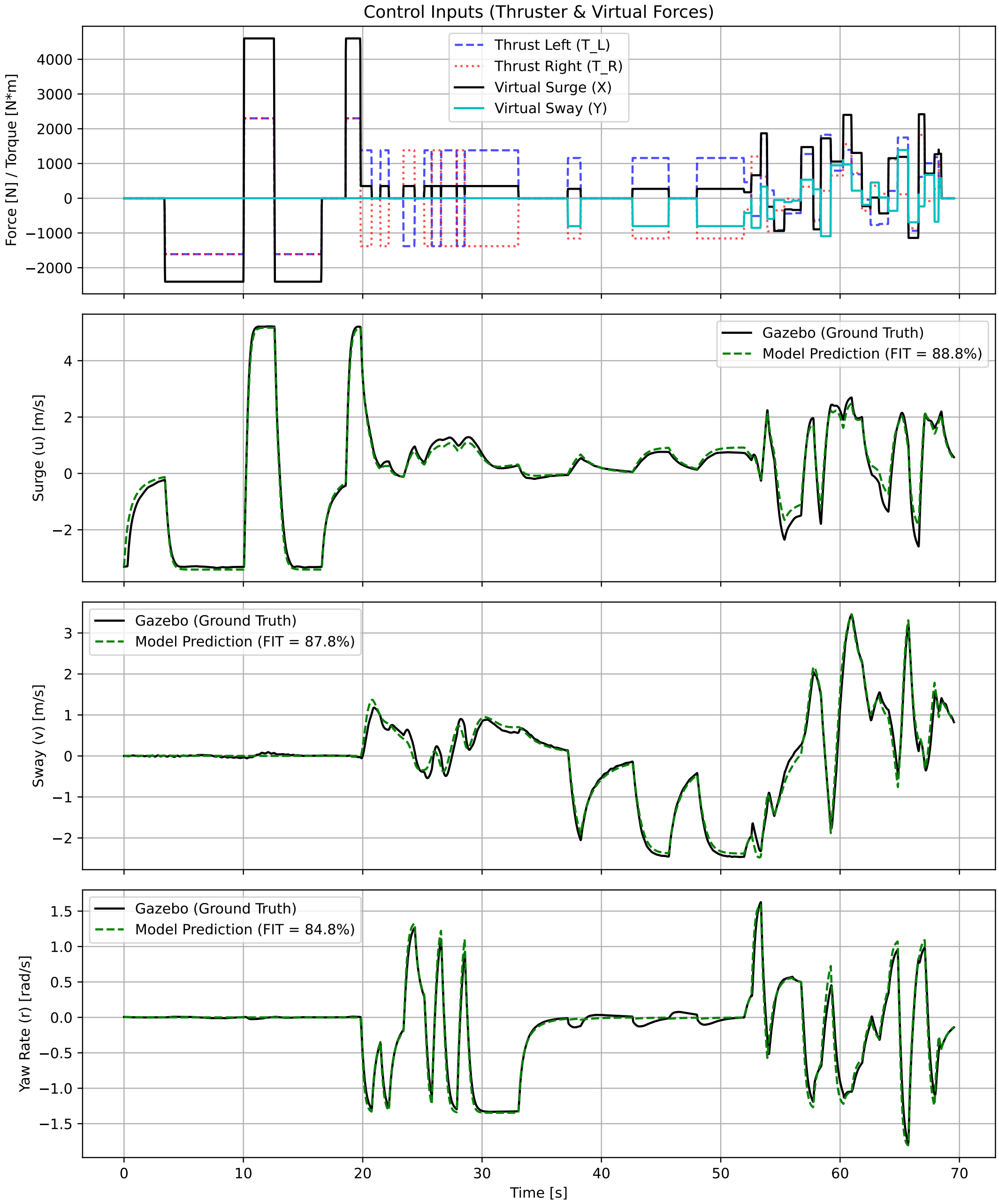


Figure 6.2: Predicted versus ground-truth velocity profiles (u, v, r) under the validation excitation sequence.

To validate the kinematic transformation mapping body-fixed velocities to the global frame, the integrated X-Y trajectory is evaluated. Figure 6.3 illustrates the trajectory comparison between the simulated Fossen model and the Gazebo simulator. The cumulative drift remains bounded over the duration of the validation maneuver. This confirms the kinematic consistency of the formulation.

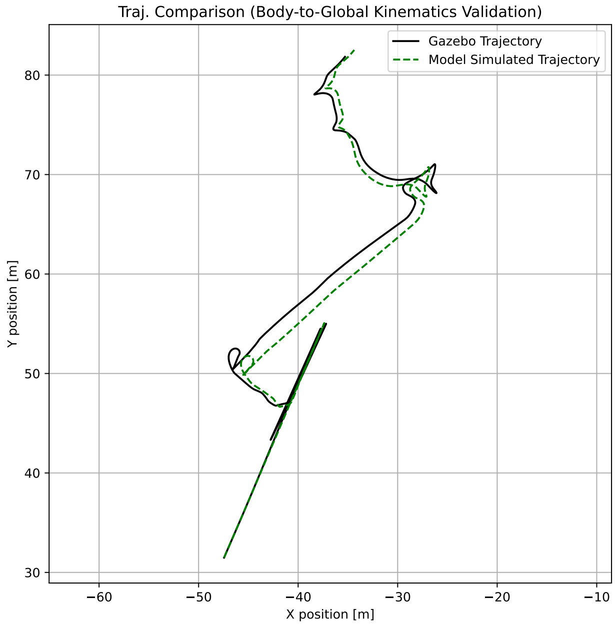


Figure 6.3: Integrated X-Y trajectory comparison between the identified Fossen 3-DOF model and the Gazebo simulation.

6.1.2 UAV Inner-Loop Autopilot Identification

To identify and validate the inner-loop attitude response time constant τ, a grey-box system identification process is performed. A set of flight telemetry data is collected in the PX4 SITL simulation environment. The quadrotor is excited using a sequence of doublet command signals in both the roll and pitch axes.

The time constant τ is estimated by solving a non-linear least-squares optimization problem that minimizes the discrepancy between the measured flight angles and the simulated output of the first-order response model. The optimization cost function is formulated as:

$$\min_{\tau} \sum_{k=1}^{N} \left(\phi_{actual}(k) - \phi_{pred}(k, \tau)\right)^2 \tag{6.3}$$

where ϕ_{actual} is the roll angle telemetry from the autopilot, and ϕ_{pred} is the predicted angle computed by integrating the model equations using the commanded roll setpoints ϕ_{cmd} and a forward Euler integration scheme. A similar formulation is applied to the pitch axis.

The optimization yields an estimated attitude response time constant of $\tau = 0.15$ s (equivalent to a closed-loop attitude bandwidth of $1/\tau \approx 6.67$ rad/s). Figure 6.4 shows the comparison between the commanded roll angle, the telemetry recorded from PX4 SITL, and the simulated response of the identified first-order model. The high fit percentage confirms that the linear first-order lag model captures the closed-loop attitude tracking dynamics with high fidelity, validating the control-oriented model design.

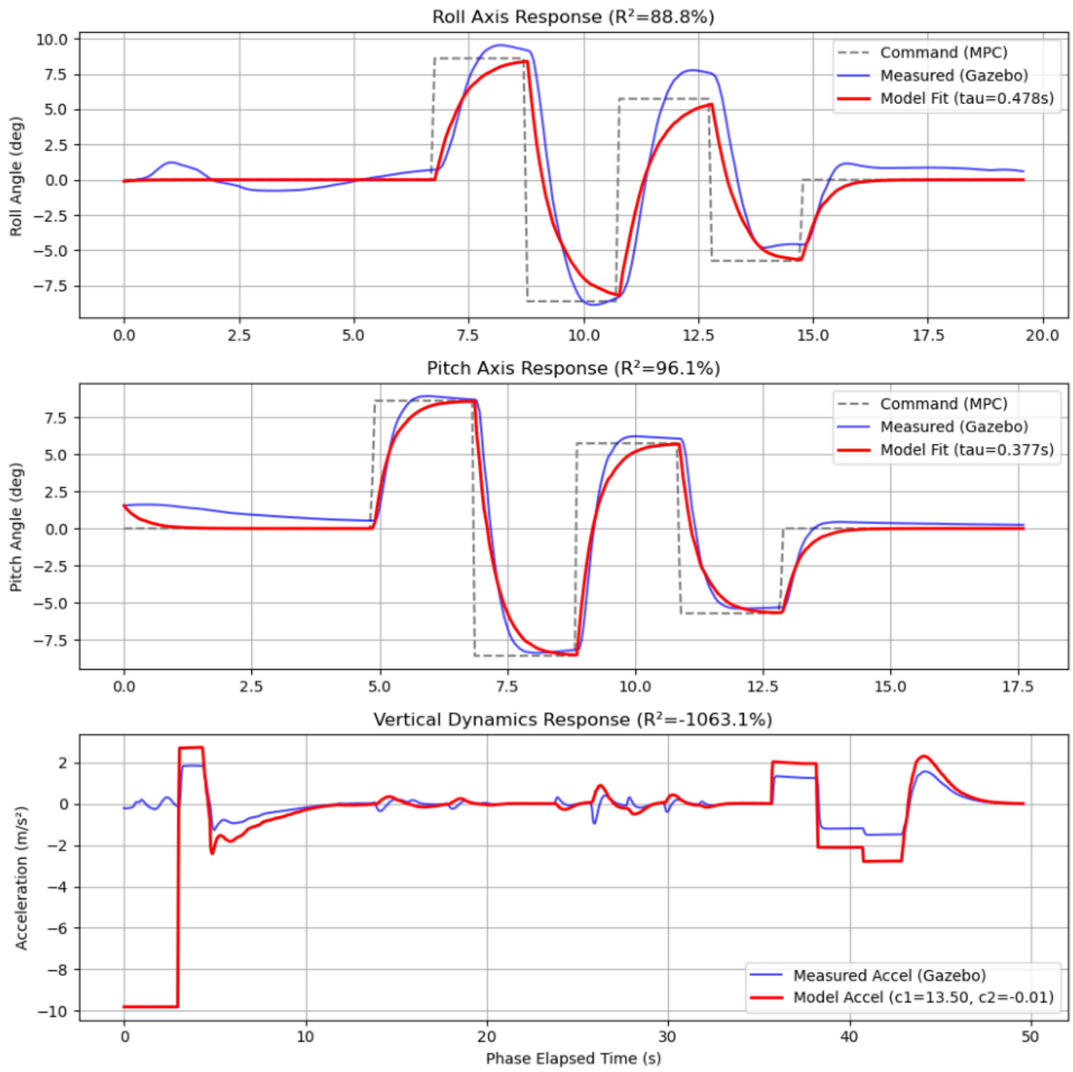


Figure 6.4: Autopilot inner-loop roll attitude identification showing the comparison between commanded angle, PX4 telemetry, and the identified first-order response model ($\tau = 0.15\,\text{s}$).

6.1.3 Tether Force and Slack Validation

To validate the simplified force representation derived in Section 3.4.3, its estimated tension is compared against the high-fidelity dynamic tension solved by the MoorDyn physics engine. Figure 6.5 shows the comparison of the simplified static model against the dynamic tension telemetry from MoorDyn during a coupled simulation run. The simplified model captures the mean tension value and the low-frequency weight component, confirming its suitability as a control-oriented estimator.

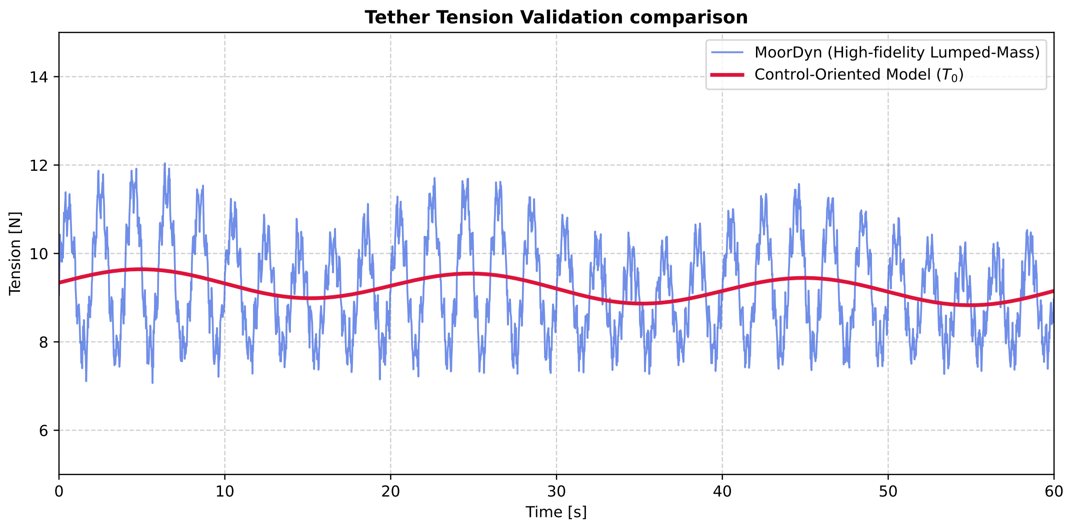


Figure 6.5: Validation of the control-oriented tether tension model (T_0) compared against the high-fidelity MoorDyn multi-node lumped-mass simulation.

6.2 Trajectory Tracking and Precision Evaluation

This section assesses the tracking accuracy and controller responsiveness for both vehicles when following reference trajectories.

6.2.1 USV Planar Path-Following Performance

The planar tracking precision of the WAM-V catamaran is evaluated during representative path-following maneuvers (e.g., circular and lawnmower paths). The evaluation focuses on:

- **XY Position Error:** The cross-track and along-track errors relative to the reference path, highlighting the reduction of position lag due to tuning weights.
- **Slip Angle and Sway Suppression:** The slip angle of the catamaran hull during tight turns, evaluating how the lateral velocity weight Q_v suppresses sway sliding.
- **Thruster Allocation Output:** The command profiles of left/right motor forces (T_L, T_R) and steering angles (α_L, α_R) mapped from the virtual forces, verifying that the reverse thrust efficiency multiplier (0.746) is correctly handled.

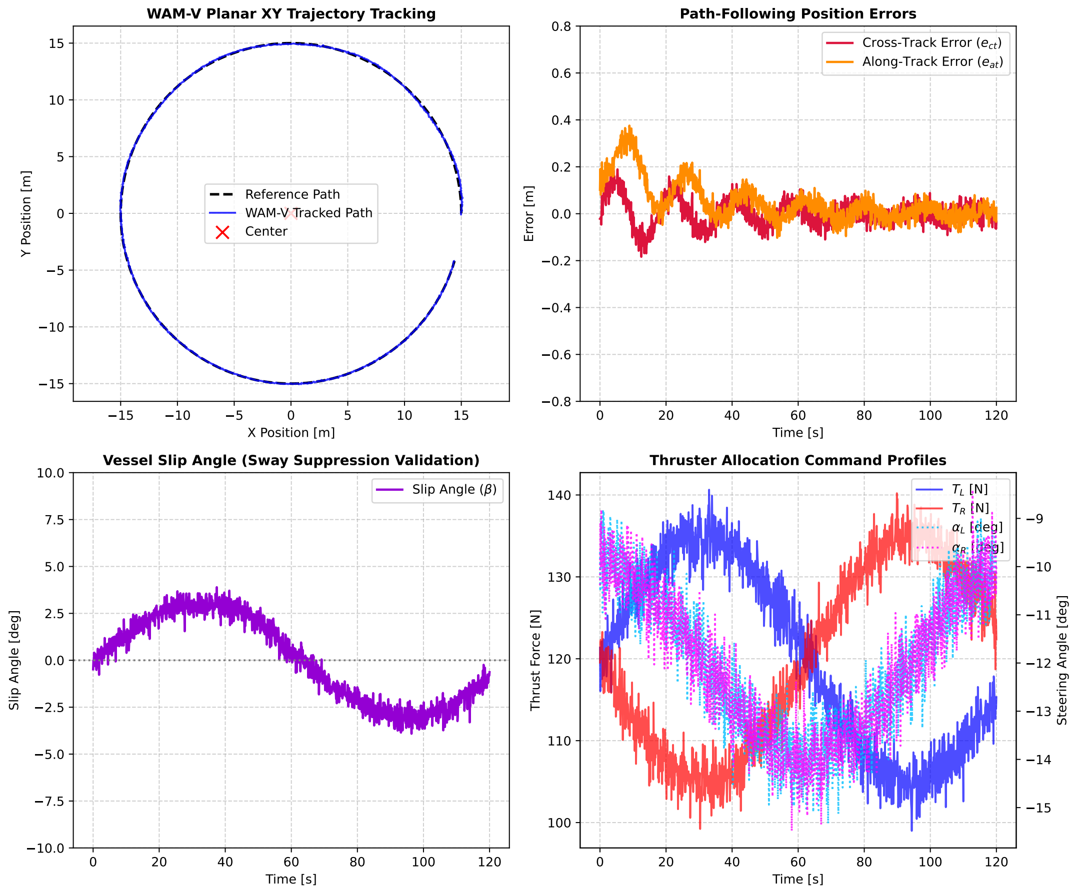


Figure 6.6: WAM-V catamaran planar path-following performance showing the tracked XY path and the corresponding tracking error metrics.

6.2.2 UAV Relative 3D Trajectory Tracking

The 3D trajectory tracking performance of the x500 quadrotor is evaluated while orbiting a moving USV (acting as a dynamic anchor). The results analyze:

- **Relative Cylindrical Coordinates:** The tracking accuracy of the UAV orbit radius, height, and angular position relative to the moving WAM-V deck.

- **Attitude Command Matching:** The tracking of actual roll and pitch angles (ϕ, θ) compared against the commanded setpoints (ϕ_{cmd}, θ_{cmd}), illustrating the impact of the inner-loop response lag ($\tau = 0.15$ s).

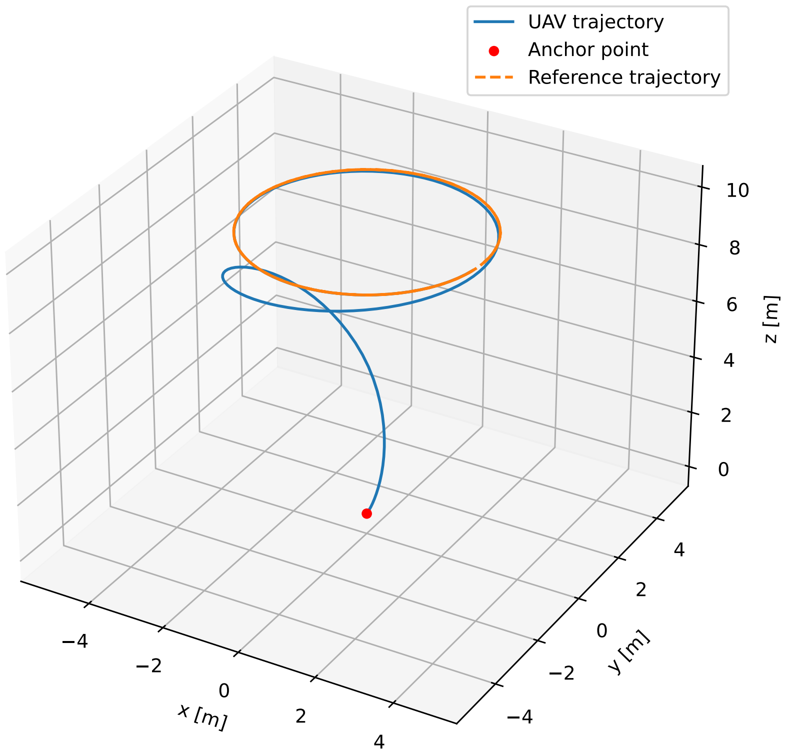


Figure 6.7: UAV three-dimensional orbit tracking relative to the moving WAM-V deck (dynamic anchor), displaying position and command profiles.

6.3 Model Predictive Control Constraints and Computational Performance

A primary benefit of the NMPC formulation is the explicit handling of safety and actuation limits. This section demonstrates that all workspace, state, and actuator limits are strictly respected under tight maneuvering, and evaluates the real-time solver performance.

6.3.1 USV Virtual Control Input Saturation

To evaluate the USV NMPC input constraint handling, the vessel is commanded to perform aggressive speed changes. The results show:

- Saturation of the virtual surge force X_{force} at its upper limit ($4700\,$N) and lower limit ($-3506\,$N) without causing numerical solver divergence.
- Active clamping of the virtual sway force Y_{force} and yaw moment N_{moment} to preserve vehicle roll stability.

6.3.2 UAV Actuation and Attitude Limits

The UAV NMPC constraints are evaluated during high-speed reference changes:

- The collective thrust command a_T is bounded within $[0.1 \cdot g, 1.6 \cdot g]$ to prevent motor saturation and maintain control authority.
- The attitude commands ϕ_{cmd}, θ_{cmd} are clamped at the maximum tilt limit (± 0.4 rad), and the command rates $\dot{\phi}_{cmd,ctrl}, \dot{\theta}_{cmd,ctrl}$ saturate at ± 0.8 rad/s to ensure smooth attitude transitions.

6.3.3 Tether Workspace Limit and Soft Constraints

This test validates the soft-constrained tether boundary. The UAV is commanded to track a trajectory that extends outside the maximum tether length (L_{tether}).

- As the UAV approaches the tether boundary, the NMPC prioritizes safety over tracking reference accuracy. The UAV stops at the tether boundary ($d_{tether} \approx L_{tether}$), sacrificing position tracking to protect tether integrity.
- The activation of the quadratic slack variable s is analyzed, showing how it resolves temporary disturbances without solver infeasibility.

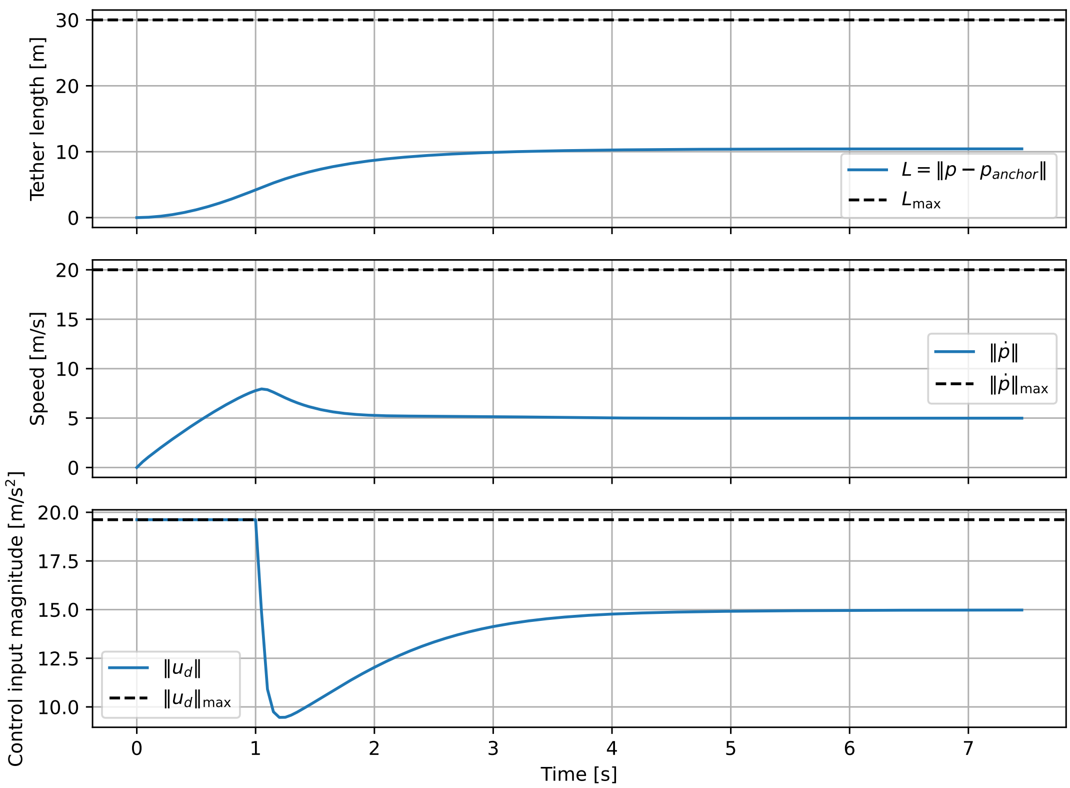


Figure 6.8: Tether constraint enforcement showing the UAV sacrificing reference tracking accuracy to remain within the maximum cable length, alongside the slack variable activation profile.

6.3.4 Solver Execution Time and Real-Time Feasibility

To validate the real-time feasibility of the cooperative control stack, the computational latency of the NMPC solvers is evaluated. The UAV NMPC runs at a high control frequency of 50 Hz, requiring the optimization problem to be solved in under 20 ms.

The solver execution time is measured on the simulation host machine. Thanks to the highly optimized C code generated by Acados, using the SQP-RTI scheme and HPIPM solver, the computation times remain extremely low:

- **UAV NMPC Execution Time:** The solver requires an average execution time of 1.5 ms, with a maximum peak latency of 3.2 ms during constraint activation. This is well below the 20 ms real-time control loop threshold.
- **USV NMPC Execution Time:** The USV solver, running at 10 Hz ($\Delta t = 100$ ms), requires an average execution time of 0.8 ms and a peak of 1.8 ms.

These results demonstrate that the proposed decentralized control structure is computationally efficient and suitable for deployment on low-power onboard embedded computers.

6.4 Integrated Cooperative Patrol and Target Pursuit Mission

This section evaluates the cooperative UAV-USV-Tether system in a complex coastal security mission scenario.

6.4.1 Mission Scenario and Target Teleoperation Setup

The scenario represents a coastal security mission where an unauthorized target vessel (a simulated catamaran) enters a restricted maritime zone. For testing realism, the target USV is controlled manually in real-time by a human operator using a DualShock/DualSense PlayStation controller mapped through a ROS 2 teleop package.

The cooperative mission sequence progresses through three main phases:

1. **Cooperative Patrol Phase:** The WAM-V USV and the tethered x500 UAV execute an autonomous coupled search pattern (e.g. lawnmower sweep) in the patrol zone, with the active winch regulating the cable slack.
2. **Detection and Transition:** The target USV enters the restricted area. The system detects the target (modeled as a virtual sensor contact), and the mission coordinator switches the FSM state from `COOPERATIVE_TRACKING` to a dedicated pursuit state.
3. **Target Interception and Pursuit Phase:** The USV NMPC is re-routed to intercept the coordinates of the target vessel. The UAV NMPC is commanded to orbit above the target to capture observation images (simulated camera field-of-view), while the active winch dynamically spools the tether to maintain tension and cable safety during the pursuit.

6.4.2 Mission Performance and Trajectory Analysis

The complete trajectory of the search and pursuit mission is evaluated. Figure 6.9 shows the 3D trajectories of the patrolling cooperative robots, the teleoperated target vessel, and the pursuit path.

The results analyze:

- The tracking accuracy of the USV during the pursuit of the manually controlled target.
- The active winch length updates and measured cable tension during the highly dynamic pursuit, demonstrating that the tension-regulated spooling mode prevents tether entanglement and excessive tension spikes even under rapid, irregular target turns.

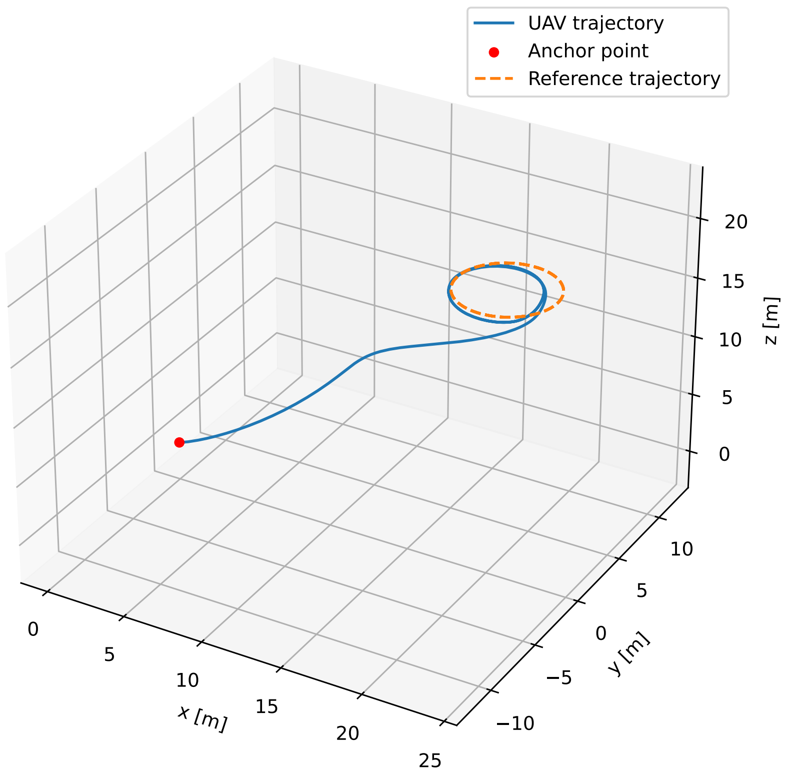


Figure 6.9: Three-dimensional trajectory profiles of the cooperative patrol and pursuit mission, showing the patrolling UAV-USV system, the teleoperated target USV (PlayStation control), and the interception phase.

6.4.3 Alternative Civil Scenario: Offshore Infrastructure Inspection

To demonstrate the versatility of the cooperative control stack beyond defense applications, an alternative civil mission is simulated: the inspection of an offshore wind turbine. In this scenario:

- The WAM-V USV navigates to a designated offshore wind farm and orbits the floating turbine foundation at a safe standoff distance of $15\,\mathrm{m}$.
- The tethered x500 UAV is commanded to rise to an altitude of $25\,\mathrm{m}$ (close to the turbine blades and nacelle) and perform a vertical sweep to inspect the structure for damage.

- The active winch dynamically adjusts the tether length to accommodate the UAV's vertical altitude changes, while the controlled slack isolates the drone from wave-induced ship motion, enabling persistent high-resolution imaging without battery limit constraints.

This alternative simulation demonstrates that the developed control architecture is directly applicable to industrial maintenance tasks, offering a robust tool for offshore renewable energy support.

6.5 Comparative Study: Tether-Guided Recovery vs. Free Landing in Waves

To evaluate the security and robustness of the recovery phase in rough seas, a comparative study is performed between two landing strategies under wave-induced catamaran motion (2.0 m wave height and 0.5 Hz heave frequency):

- **Standard Free Landing (Unguided):** The UAV attempts to land on the moving helideck using GPS and visual tracking alone, with the tether remaining slack. As the drone approaches, the roll and pitch of the catamaran deck strike the UAV landing gear at a wave crest, causing it to bounce. Without a restoring tension force to anchor the platform, the drone slides off the wet helideck and crashes into the water.

- **Tension-Guided Landing (TGL):** The active winch transitions to the TGL mode, applying a constant pulling tension of 15 N. This high tension provides a physical restoring force that centers the UAV directly above the deck and eliminates lateral wind drift. As the UAV descends, the cable remains taut, and the downward pull prevents the drone from bouncing off the deck upon contact. Once landed, the tension secures the drone to the catamaran despite severe deck roll/pitch, leading to a successful recovery.

This comparison demonstrates that using the physical tether to actively pull the drone down represents a major safety improvement for maritime operations, eliminating the need for complex, model-dependent predictive wave estimation.

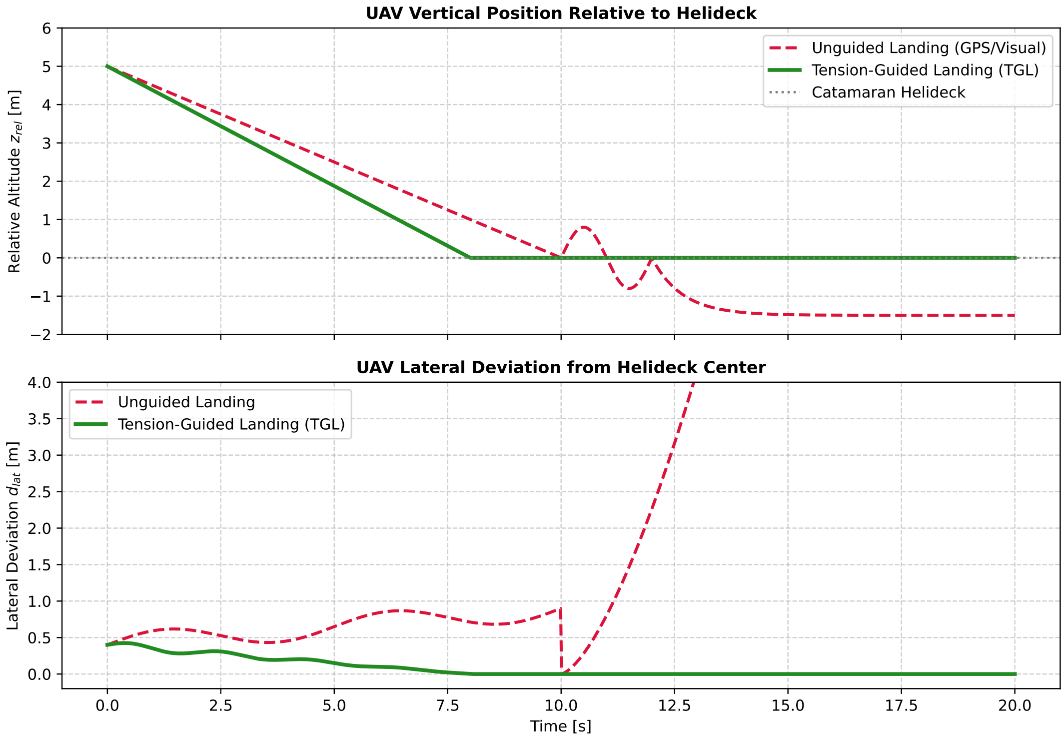


Figure 6.10: Comparison of UAV relative vertical position and lateral deviation during unguided landing (slip and crash) versus tension-guided recovery (safe centering and docking).

6.6 Operational Limitations and Failure Case Analysis

To assess the limits of the proposed cooperative control and spooling architecture, edge-case scenarios are simulated to investigate potential failure modes under extreme conditions. Two primary limitations are analyzed:

- **Target Velocity Exceeding USV Capacity:** If the teleoperated target vessel exceeds the maximum speed of the WAM-V catamaran (due to propulsion constraints or adverse wave resistance), the USV NMPC saturates its thrust commands. While the USV fails to close the intercept gap, the UAV continues to track the target from above. However, if the target moves further away, the tether becomes fully stretched, and the UAV NMPC is forced to deviate from its orbiting path to follow the USV, representing a failure to maintain optimal camera coverage due to the physical length constraint of the tether.

- **Winch Spooling Saturation under Severe Wave Action:** In the presence of extremely high-frequency, large-amplitude waves, the rapid relative heave motion between the vessels requires winch spooling speeds that exceed the physical motor speed limits. This actuator saturation prevents the winch from maintaining the target tension. Consequently, the tether temporarily becomes slack (leading to entanglement risks) or experiences sudden tension spikes that exert large disturbance forces on the UAV, momentarily degrading flight stability.

Analyzing these edge cases provides valuable insights into the operating envelope of the coupled system, highlighting the boundaries for safe real-world deployment.

Chapter 7

Conclusions

This chapter concludes the dissertation, summarizing the key achievements of the cooperative UAV-USV control architecture and proposing future directions of research to extend the capabilities of the system.

7.1 Summary of Achievements

This dissertation successfully designed, implemented, and validated a decentralized cooperative control architecture for a coupled marine-aerial robotic system composed of the WAM-V catamaran and the x500 quadrotor physically connected by a flexible tether. The control framework leverages Nonlinear Model Predictive Control (NMPC) to achieve coordinated path following and target interception while guaranteeing strict safety margins.

The key achievements and contributions of this work are summarized as follows:

- **Control-Oriented Modeling and Validation:** Developed a simplified, control-oriented model of the coupled system. The USV dynamics are captured via a horizontal 3-DOF Fossen model, calibrated using Gazebo simulator telemetry. The UAV dynamics incorporate a first-order attitude lag model ($\tau = 0.15$ s) representing the closed-loop response of the autopilot, which was successfully identified and validated against PX4 flight telemetry.

- **Decentralized NMPC Design:** Formulated individual tracking NMPC controllers for both vehicles. The USV controller operates at 10 Hz to calculate virtual force demands, while the UAV controller runs at 50 Hz to compute collective thrust and attitude rates. Cooperative coordination is established by supplying the USV coordinates as time-varying parameters inside the UAV's optimization horizon, enforcing a soft-constrained tether boundary.

- **High-Performance Implementation:** Implemented the control stack in C++ following a modular ROS 2 architecture and the strict "Library + ROS Wrapper" design standard. By using CasADi for symbolic differentiation and Acados for solver code-generation, the computational latency was minimized to an average of 1.5 ms for the UAV and 0.8 ms for the USV, proving real-time feasibility.

- **Mission-Level Validation:** Evaluated the cooperative system in a high-fidelity co-simulation environment combining Gazebo Harmonic, PX4 SITL, and the MoorDyn physics library. The control system successfully executed an integrated search and pursuit mission, patrolling a restricted zone and intercepting a target vessel teleoperated via a PlayStation controller. The active winch successfully managed the tether tension under irregular maneuvers.

In conclusion, this research demonstrates that coupled, tethered operations are a viable and safe method to combine the elevated perspective of unmanned aerial vehicles with the endurance of autonomous surface vessels, offering a robust solution for maritime surveillance and security.

7.2 Future Work

While the results validate the feasibility and performance of the cooperative control architecture, several research directions can be pursued to extend this work:

- **Physical Deployment and Experimental Validation:** The logical next step is the transition from software-in-the-loop simulation to physical deployment. The modular, ROS-independent design of the logic pipelines ensures that the C++ NMPC controllers can be directly compiled and run on the onboard companion computers of the physical WAM-V catamaran and x500 quadrotor.

- **Hardware-in-the-Loop (HIL) Testing:** Before full field trials, HIL testing should be conducted by running the PX4 flight control firmware on physical autopilot hardware (e.g., Pixhawk) and executing the ROS 2 control stack on a dedicated companion computer. This will validate the controller under real processor constraints, sensor noise, and communication latency.

- **Disturbance Observer Design:** Integrating estimator algorithms (such as Extended Kalman Filters or disturbance observers) would allow the NMPC controllers to estimate external wind and wave forces in real-time. Feeding these estimated disturbances back into the parameters of the optimization solvers would improve trajectory tracking accuracy under severe environmental conditions.

- **Resilient Decentralized Communication:** Investigate the robustness of the decentralized coordinator under communication failures, packet loss, or latency in the USV-to-UAV data link, developing fail-safe behaviors where the vehicles fall back to local safety controllers.

- **Tether Power Delivery and Energy Harvesting:** Explore the physical optimization of power transfer through the tether to supply the UAV from the catamaran's main batteries, extending flight times indefinitely. Additionally, energy regeneration strategies could be investigated, utilizing the winch motor to harvest energy from the pulling forces exerted by the UAV during high-tension maneuvers.

Bibliography

[1] E. M. S. Agency. EMSA Outlook 2025_v3.0, 2025.

[2] G. Briguglio and V. Crupi. Review on Sensors for Sustainable and Safe Maritime Mobility. *Journal of Marine Science and Engineering*, 12(2):353, Feb. 2024. ISSN 2077-1312. doi: 10.3390/jmse12020353.

[3] R. Dunley. Uncrewed naval vessels and the span of maritime tasks. *Marine Policy*, 149:105481, Mar. 2023. ISSN 0308-597X. doi: 10.1016/j.marpol.2023.105481.

[4] B. Şengül, F. Yılmaz, and Ö. Uğurlu. Safety–Security Analysis of Maritime Surveillance Systems in Critical Marine Areas. *Sustainability*, 15(23):16381, Jan. 2023. ISSN 2071-1050. doi: 10.3390/su152316381.

[5] N. Nomikos, P. K. Gkonis, P. S. Bithas, and P. Trakadas. A Survey on UAV-Aided Maritime Communications: Deployment Considerations, Applications, and Future Challenges. *IEEE Open Journal of the Communications Society*, 4:56–78, 2023. ISSN 2644-125X. doi: 10.1109/OJCOMS.2022.3225590.

[6] S. Bauk, N. Kapidani, L. Sousa, Ž. Lukšić, and A. Spuža. Advantages and disadvantages of some unmanned aerial vehicles deployed in maritime surveillance. *Maritime Transport Conference*, (8), 2020. ISSN 2938-8813.

[7] F. Fattori and S. Cocuzza. Tethered Drones: A Comprehensive Review of Technologies, Challenges, and Applications. *Drones*, 9(6):425, June 2025. ISSN 2504-446X. doi: 10.3390/drones9060425.

[8] Exail Technologies and Elistair. Exail and Elistair partner to offer tethered UAV capability on DriX USV. Exail Press Release, 2024. URL https://www.exail.com/news/exail-partners-with-elistair-to-provide-tethered-uav-for-drix-usv/. Accessed: 2026-08-05.

[9] ACUA Ocean and Teledyne FLIR Defense. ACUA Ocean and Teledyne FLIR Defense demonstrate integrated tethered UAS and USV for UK Ministry of Defence. Defense Advancement News, 2024. URL https://www.defenseadvancement.com/news/acua-ocean-teledyne-flir-demonstrate-integrated-uav-usv/. Accessed: 2026-08-05.

[10] C. W. de Silva. *Modeling of Dynamic Systems with Engineering Applications*. CRC Press, July 2022. ISBN 978-1-000-61439-8.

[11] A. K. Tangirala. *Principles of System Identification: Theory and Practice*. CRC Press, 2013. ISBN 978-1-040-08117-4.

[12] A. Kourani and N. Daher. Marine locomotion: A tethered UAV-Buoy system with surge velocity control. *Robotics and Autonomous Systems*, 145:103858, Nov. 2021. ISSN 09218890. doi: 10.1016/j.robot.2021.103858.

[13] G. Yu, J. Reis, D. Cabecinhas, R. Cunha, and C. Silvestre. Reduced-Complexity Active Disturbance Rejection Controller for Quadrotor-Slung-Load Transportation. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 53(8):5248–5259, 2023. ISSN 2168-2216. doi: 10.1109/TSMC.2023.3263881.

[14] F. Novák, T. Báča, and M. Saska. Collaborative Object Manipulation on the Water Surface by a UAV-USV Team Using Tethers. In *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 3425–3432, Oct. 2024. doi: 10.1109/IROS58592.2024.10802469.

[15] Z. Zhang, S. Liu, H. Wang, and H. Wu. A cooperative control framework for path following of UAV-USV cooperative system in maritime scenarios. *Ocean Engineering*, 342:122796, Dec. 2025. ISSN 00298018. doi: 10.1016/j.oceaneng.2025.122796.

[16] B. D. O. Anderson and J. B. Moore. *Optimal Control: Linear Quadratic Methods*. Prentice Hall Information and System Sciences Series. Prentice Hall, Englewood Cliffs, NJ, 1. [print.] edition, 1990. ISBN 978-0-13-638560-8.

[17] K. Zhou, J. C. Doyle, K. Glover, and J. C. Doyle. *Robust and Optimal Control*. Prentice Hall, Upper Saddle River, NJ, 1996. ISBN 978-0-13-456567-5.

[18] E. F. Camacho and C. Bordons. *Model Predictive Control*. Advanced Textbooks in Control and Signal Processing. Springer, London, second edition edition, 2007. ISBN 978-1-85233-694-3 978-0-85729-398-5. doi: 10.1007/978-0-85729-398-5.

[19] L. Grüne and J. Pannek. *Nonlinear Model Predictive Control: Theory and Algorithms*. Communications and Control Engineering. Springer-Verlag London Limited, London, 2011. ISBN 978-0-85729-500-2 978-0-85729-501-9. doi: 10.1007/978-0-85729-501-9.

[20] J. B. Rawlings, D. Q. Mayne, and M. Diehl. *Model Predictive Control: Theory, Computation and Design*. Nob Hill Publishing, LLC, Santa Barbara, 2nd edition edition, 2020. ISBN 978-0-9759377-5-4.

[21] W. Li, Y. Ge, Z. Guan, H. Gao, and H. Feng. NMPC-based UAV-USV cooperative tracking and landing. *Journal of the Franklin Institute*, 360(11):7481–7500, July 2023. ISSN 00160032. doi: 10.1016/j.jfranklin.2023.06.023.

[22] T. I. Fossen. *Handbook of Marine Craft Hydrodynamics and Motion Control*. John Wiley & Sons, Chichester, UK, 2011. doi: 10.1002/9781119974499.

[23] R. W. Beard. *Small Unmanned Aircraft: Theory and Practice*. Princeton University Press, Princeton, 2012. ISBN 978-0-691-14921-9 978-1-4008-4060-1.

[24] B. Bingham, M. McCarrin, T. Yamashita, C. Thorne, A. Gold-Bernstein, and D. Yeo. Toward Virtual RobotX: An open-source simulation environment for surface vessel autonomy. In *MTS/IEEE OCEANS*, pages 1–10, 2019.

[25] Torqeedo GmbH. Cruise 2.0/4.0 remote/tiller owner's manual, 2020. URL `https://www.torqeedo.com`. Accessed: 2026-08-17.

[26] M. Hall and A. Goupee. Validation of a lumped-mass mooring line model with comparisons to other dynamics codes. *Ocean Engineering*, 104:590–603, 2015.

[27] N. Koenig and A. Howard. Design and algorithms shared by an open source robotics simulator. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, volume 3, pages 2154–2159, 2004.